# The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

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## Abstract

**Objective.** Temporal-interference (TI) stimulation applies two high-frequency currents whose amplitudes beat at a low difference frequency, offering focal, steerable stimulation at depth. But the field carries no power at the beat, so no passive linear element can follow it; recovering it needs a nonlinearity, generally sought in single-cell ion channels. We ask how much of demodulation and its tuning belongs instead to the neural population. **Approach.** We treat the firing-rate nonlinearity of a neural mass as an amplitude detector and its recurrent network near a Hopf bifurcation as a resonant amplifier, across a modeling ladder: a Jansen–Rit column, a laminar model, an exact next-generation mean field, and its underlying quadratic integrate-and-fire network. **Main results.** Transfer-function curvature acts as a square-law detector recovering the beat, and the network amplifies it resonantly at its natural frequency, the gain following $1/\gamma$ as the damping vanishes and tunable by connectivity at fixed detection. We offer detection as a *candidate*: square-law transduction is generic for any smooth weak nonlinearity, but the sigmoid family's efficiency ceiling is reached only in a threshold limit whose microvolt transfer width lies far below any realistic population transfer. Amplification is independent of that question, since the measured gain is normalized by the detector output. The account reproduces TI's known behavior: inert-carrier independence at matched polarization, a peak when the beat matches a region's rhythm, and spike-timing realignment with little rate change, as seen in vivo. **Significance.** Tuning is a population property whatever detects. TI efficacy should thus be as much a property of the brain as of the device, largest near a bifurcation and set by state and connectivity. The column is a tuned receiver whose detector may sit in the single cell, which reframes dose optimization around the target's dynamics and yields a measurement deciding the detector class.

**Keywords:** temporal interference; non-invasive neuromodulation; transcranial stimulation; neural mass model; amplitude demodulation; Hopf bifurcation; cross-frequency coupling; criticality.

## Contents

## 1 Introduction

Reaching a deep brain target without stimulating the tissue above it is a long-standing goal of non-invasive neuromodulation, and temporal-interference (TI) stimulation [1] is its most prominent recent proposal. TI applies two sinusoids at $f_1$ and $f_2 = f_1 + \Delta f$ through separate electrode pairs (typically $f_1 \sim 1$–2 kHz, $\Delta f \sim 1$–100 Hz). Where the two fields overlap, their superposition is an AM signal whose carrier is near $f_1$ and whose envelope beats at the difference frequency $\Delta f$. The appeal is spatial: the envelope amplitude peaks in a focal, steerable region at depth, while superficial tissue sees only the (putatively inert) kHz carrier—a promise of non-invasive, focal, deep stimulation that explains the intense interest.

The difficulty is conceptual rather than technological. Write the field as

$$s(t) = \varepsilon\,[\,1 + m\cos(\Omega t)\,]\cos(\omega_c t), \qquad \omega_c = 2\pi f_c, \quad \Omega = 2\pi\,\Delta f, \tag{1}$$

where $m$ is the modulation depth, $\Delta f$ the envelope (beat) frequency, $f_c$ the carrier frequency (the mean $(f_1 + f_2)/2$ of the two applied tones, the envelope beating at their difference $\Delta f = f_2 - f_1$), and $\varepsilon$ the field amplitude (refined to the post-coupling polarization in §2.2); its spectrum has support *only* at $\omega_c$ and $\omega_c \pm \Omega$ (Fig. 3): there is no energy at the envelope frequency $\Omega$. A linear, time-invariant (LTI) system maps a sinusoid to a sinusoid of the same frequency; it can attenuate or phase-shift the three carrier lines but can *never* synthesize a new line at $\Omega$. The membrane's passive RC low-pass—often invoked as "the neuron follows the slow envelope"—therefore cannot by itself demodulate TI. Demodulation requires a nonlinearity. This is the founding lesson of AM radio [2]: a receiver cannot recover the message with linear components alone; it needs a nonlinear element (a square-law device or a diode) to shift spectral energy down to *baseband*—the low-frequency band near 0 Hz where the message lives—followed by a tuned filter to select the station and a low-pass to read the envelope.

Where, then, is the nonlinearity in the brain? The original passive-filtering intuition is now widely regarded as insufficient, since a linear filter cannot create the envelope line [3]. The leading biophysical account places the nonlinearity in single-neuron ion channels: Mirzakhalili et al. [3] show in Hodgkin–Huxley/Frankenhaeuser–Huxley axon models that TI works through nonlinear, ion-channel-mediated rectification, in which asymmetric voltage-gated conductances rectify the carrier and, after membrane low-pass, follow the envelope—the textbook rectify-then-filter detector. Other studies report that TI thresholds inherit the carrier's frequency dependence, suggesting a mechanism shared with direct kHz stimulation (conduction block, onset response) rather than a dedicated envelope channel [4, 5], and in some peripheral preparations the response is not driven by envelope extraction at all [5]. In vivo, TI reliably alters spike *timing* but not firing rate in the primate brain, and is much weaker than conventional low-frequency stimulation under human-relevant conditions [6]. Most tellingly, recent slice work finds that pyramidal and parvalbumin (PV) interneurons respond differently and that most pyramidal TI responses disappear when synaptic transmission is blocked, leading Caldas-Martinez et al. to conclude that TI is "largely a network phenomenon" [7]; computational network models point the same way [8].

Here we ask how far a population-scale mechanism, built from two ingredients already in standard cortical models, can go: a static firing-rate *sigmoid*, whose nonzero curvature $\sigma''$ makes it a square-law demodulator, and a *second-order-synapse network* whose synaptic kinetics filter the carrier away while a weakly-damped *stable focus* resonantly amplifies the recovered beat component at the network's natural frequency (Fig. 1). The amplification is a property of the focus, not of the Hopf specifically: it sharpens as the focus nears criticality, which the column approaches near *both* the saddle-node on an invariant circle (SNIC, a slower rhythm) and the Hopf (alpha). Where existing accounts locate the nonlinearity in cellular excitability, ours reads it out at the mesoscopic scale: the sigmoid is a coarse-grained summary of the population's single-cell input–output curves, but once coarse-grained it is a *distinct mesoscopic nonlinearity* with enough curvature to detect envelopes, whatever microscopic mechanism supplies it. This is a candidate network-level

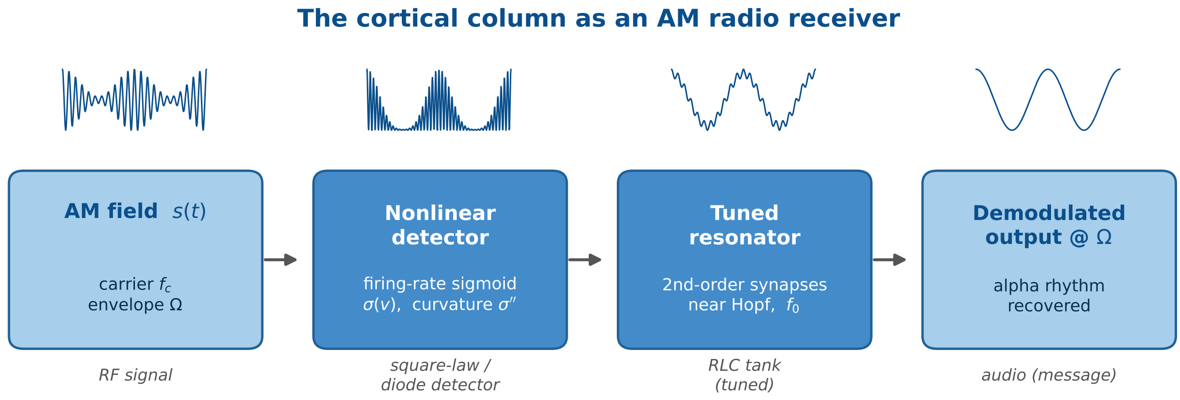


**Figure 1: The cortical column as an AM radio receiver (schematic).** An AM field (carrier $f_c$, envelope $\Omega$) reaches a nonlinear detector (the firing-rate sigmoid, curvature $\sigma''$), whose output passes through a tuned resonator (the second-order-synapse network near a Hopf, natural frequency $f_0$), recovering a beat-frequency component as an oscillation at $\Omega$. The analogy is functional rather than circuit-identical. In a classic AM receiver the tuned resistor–inductor–capacitor tank sits *before* the detector and selects the carrier [2]; here the sigmoid generates the baseband beat component, and a *baseband* resonator downstream of it selects and amplifies modulation frequencies near the column's own rhythm, so TI "broadcasts" to whichever columns are tuned to its beat frequency. A carrier-selective stage upstream of the detector does appear in the laminar model (§4.5).

route to TI demodulation—carrier-independent (at fixed polarization) and frequency-selective—complementary to the disinhibitory network effect (PV-interneuron silencing) reported by [7]; both place TI's action in the population rather than the single cell. We develop it first in a single Jansen–Rit column, analyzable in closed form, then in the two-band laminar model (LaNMM), whose own alpha and gamma loops bring it closer to cortex, and finally in an exact next-generation mean field, where the demodulating nonlinearity is derived rather than assumed.

This work builds on a line of research in which neural-mass models implement AM-radio principles *internally*, for the brain's own signals. In our cross-frequency-coupling account of predictive coding [9], the LaNMM realizes a columnar Comparator through sigmoid-enabled signal-to-envelope (SEC) and envelope-to-envelope (EEC) couplings, where error evaluation is literally demodulation—the sign-reversed prediction "un-modulates" the carrier. The Hierarchical Amplitude Modulation (HAM) framework [10] generalizes this to nested envelopes with constant-$Q$ (log-spaced) bands, a candidate origin of $1/f$ spectra—again through the sigmoid's quadratic cross-term. The present work turns this machinery outward: if the cortex is an AM transceiver for its own rhythms, it can also be an AM receiver for an external carrier.

## 2 Theory

### 2.1 Neural mass models and the Jansen–Rit circuit

A neural mass model (NMM) collapses the activity of a large population ($\sim 10^4$) of similar neurons into a handful of population-averaged variables [11, 12]. Two elements recur in every such model. First, each synapse acts as a *linear filter* $\hat{K}$ that converts an incoming presynaptic firing rate $r(t)$ into a postsynaptic potential (PSP) $x(t) = \hat{K}[r]$; equivalently $r = \hat{L}[x]$ with $\hat{L} = \hat{K}^{-1}$ a differential operator whose few coefficients set the synapse's gain, decay time and delay [11]. In the Jansen–Rit (JR) family the kernel is the "alpha function" $h(t) = Aa\,t\,e^{-at}$ (excitatory; $B, b$ for the inhibitory synapse), i.e. the second-order operator $\hat{L} = (Aa)^{-1}(\mathrm{d}/\mathrm{d}t + a)^2$, so each PSP obeys $\ddot{x} + 2a\dot{x} + a^2 x = Aa\,r(t)$. Second, the PSPs converging on a population sum to a mean membrane potential $v = \sum_s x_s$, which a static, saturating *transfer function*—the sigmoid—turns back into a population firing rate [13],

$$\sigma(v) = \frac{2e_0}{1 + e^{\rho(v_0 - v)}}, \tag{2}$$

with maximal firing rate $2e_0$, half-maximal at the inflection $v_0$, and slope set by $\rho$; it saturates because firing rates are bounded [11]. The sigmoid is the only nonlinearity in the model, and—as we show—it is what makes (de)modulation possible. Its curvature, which does the demodulating, originates in the single neuron's own input–output nonlinearity: $\sigma$ is the mean-field (ensemble) average of single-neuron transfer curves over a heterogeneous, noisy population, so $\sigma''$ is inherited from single cells and only reshaped—smoothed and broadened—by the spread of states, thresholds and noise: the curvature is cellular in origin, but its demodulation is read out at the population level by the generic sigmoid, while the network supplies the resonant *amplification* we use here (§2.4). Frequency-selective resonance can also be intrinsic to single cells (e.g. $I_h$ and $I_M$ currents [14]); we locate it instead in the network, where second-order synapses near criticality generate it even when the cells do not [15, 16].

This becomes exact one rung up the modeling ladder [11]. For a network of all-to-all coupled quadratic integrate-and-fire (QIF) neurons with Lorentzian-distributed excitabilities, the Ott–Antonsen ansatz yields a *closed, exact* mean field for the population firing rate $r$ and mean potential $v$—the next-generation (Montbrió–Pazó–Roxin, MPR) mass [17], completed with the second-order synapses of neural-mass models in the exact "NMM2" theory [18, 19]:

$$\dot{r} = \frac{\Delta}{\pi} + 2rv, \qquad \dot{v} = v^2 + \bar{\eta} - \pi^2 r^2 + I(t), \tag{3}$$

with the applied current $I(t)$ entering linearly, the common background excitability $\bar{\eta}$ (the bifurcation parameter), and the heterogeneity half-width $\Delta$ setting the smoothing. Here the demodulating nonlinearity is *derived, not postulated*: the quadratic $v^2$ term (with the $rv$ coupling) is an exact square-law rectifier of the input, inherited directly from the single-neuron QIF dynamics. The Jansen–Rit sigmoid is the heuristic (NMM1) stand-in for this transfer—exact in the slow-synapse limit [19]—so our square-law result is a generic property of the mean field, not an artifact of expanding one postulated curve. The same exact theory already exhibits weak-field resonance and Arnold tongues that sharpen with self-coupling [19, 18], the near-Hopf amplification we exploit below.

*What the transfer functions represent.* In every model the demodulating object is a *population* input–output map rather than a passive somatic membrane element. In JR/LaNMM it is the sigmoid (2), the effective transfer of an excitable neuronal ensemble (mean PSP in, population firing rate out); in NMM2 the same role is played by the exact MPR quadratic (3). Its curvature is the mesoscopic trace of cellular nonlinearities (thresholds, spike generation, refractoriness, noise, saturation), so reading the demodulation off $\sigma''$, or off the derived $v^2$ term, is legitimate at this scale. The membrane low-pass does not contradict this: it bears on *access*, not legitimacy. The open biophysical question is which fast compartment (axon initial segment, node, terminal, or channel) lets a kHz carrier reach that nonlinear transfer (§2.6); once the carrier is sampled there, the neural mass describes the population-level demodulation and near-Hopf amplification of the recovered beat component.

The single-column JR circuit couples a pyramidal population to one excitatory and one inhibitory interneuron population through this synapse+sigmoid motif. Writing $y_0$ for the excitatory PSP driving the interneurons, $y_1$ and $y_2$ for the excitatory and inhibitory PSPs onto the pyramidal population, and the local-field-potential (LFP) proxy $v = y_1 - y_2$, with external input $p$,

$$\ddot{y}_0 = Aa\,\sigma(y_1 - y_2 + s(t)) - 2a\dot{y}_0 - a^2 y_0, \tag{4}$$

$$\ddot{y}_1 = Aa\left[\,p + C_2\,\sigma(C_1 y_0)\,\right] - 2a\dot{y}_1 - a^2 y_1, \tag{5}$$

$$\ddot{y}_2 = Bb\,C_4\,\sigma(C_3 y_0) - 2b\dot{y}_2 - b^2 y_2, \tag{6}$$

**Table 1:** Jansen–Rit parameters [12].

| Symbol | Meaning | Value |
|---|---|---|
| $A$, $B$ | excit./inhib. PSP gain | 3.25, 22 mV |
| $a$, $b$ | synaptic rates | 100, 50 $\mathrm{s}^{-1}$ |
| $v_0$, $e_0$, $\rho$ | sigmoid inflection, max rate/2, slope | 6 mV, 2.5 $\mathrm{s}^{-1}$, 0.56 $\mathrm{mV}^{-1}$ |
| $C$ | global connectivity | 135 |
| $C_1, C_2, C_3, C_4$ | couplings | $C$, $0.8C$, $0.25C$, $0.25C$ |
| $p$ | external input (bifurcation knob) | 0–400 Hz |

integrated as six first-order ODEs with the parameters of Table 1 [12]. The term $s(t)$ is the applied-field perturbation, introduced next; the read-out is $v = y_1 - y_2$.

For the biophysical realization (§4.5) we use a second model built from the same ingredients: the laminar neural mass model (LaNMM) [9, 20], which embeds this synapse+sigmoid motif in a two-band cortical column—a deep Jansen–Rit-like subcircuit (pyramidal $P_1$ with spiny-stellate (SS) and somatostatin (SST) interneurons; intrinsic alpha, ∼10 Hz) coupled to a superficial PING-like (pyramidal–interneuron network gamma) subcircuit (pyramidal $P_2$ with PV interneurons; intrinsic gamma, ∼40 Hz). The single JR column lets us isolate and analyze the mechanism; the LaNMM then exhibits it in a column with its own fast/slow loops and two native resonances. Everything below applies to both: the field couples through a sigmoid, and the recovered beat component drives a resonant circuit.

### 2.2 From applied field to a squared-envelope beat component

We model the applied TI field $E(t)$ (V/m) as a quasi-static perturbation of the membrane potential that the spike-generating nonlinearity reads. "Quasi-static" means we neglect electromagnetic propagation and induction (wavelengths ≫ head, frequencies far below tissue relaxation), so the field acts as an instantaneous polarization; taking it to add directly inside the sigmoid argument,

$$\varphi(t) = \sigma\big(v(t) + \lambda\, E(t)\big), \qquad E(t) = E_0\,[\,1 + m\cos\Omega t\,]\cos\omega_c t \tag{7}$$

This is the standard weak-field coupling $\delta V_m = \vec{\lambda}\cdot\vec{E}$ of transcranial-stimulation modeling [11], with $\lambda$ a lumped polarization length. Three quantities must be kept distinct: the applied field $E(t)$, the induced membrane polarization $\delta V_m(t)$ *after* cable/membrane filtering, and the argument of the firing-rate map. The chain is

$$E(t) \xrightarrow[\text{cable/membrane}]{} \delta V_m(t) = \kappa(\mathbf{x}, \theta, f_c)\, E(t) \xrightarrow[\text{nonlinearity}]{} \sigma\big(v + \delta V_m\big) \xrightarrow[\text{network}]{} \text{resonance}, \tag{8}$$

with coupling $\kappa(f_c) = \lambda\,|H_m(f_c)|$ (mV per V/m) lumping cell geometry, orientation and the effective polarization length $\lambda$ with the membrane band-pass $H_m$ (§2.6). We therefore define $\varepsilon$ as the *post-coupling polarization* amplitude that actually reaches the nonlinearity, $\varepsilon \equiv \kappa(f_c)E_0 = \lambda|H_m(f_c)|E_0$, with $s(t) \equiv \delta V_m(t)$, recovering Eq. (1)—not the raw applied field. In the main text we take the favorable quasi-static, direct limit $\kappa = \lambda$ ($|H_m| \to 1$), which isolates the sigmoid+resonance principle and is appropriate when a fast nonlinearity (axonal/nodal/terminal) samples the carrier before RC attenuation; §2.6 restores $H_m(f_c)$ and shows that the carrier dependence enters quadratically, $A_\Omega \propto |H_m(f_c)|^2$.

Expanding $\sigma$ about an operating point $v^*$ (a Volterra/Taylor expansion, as in [10]) gives $\sigma(v^*{+}s) \approx \sigma(v^*) + \sigma'(v^*)\, s + \frac{1}{2}\sigma''(v^*)\, s^2 + \dots$ The truncation holds only while $|s|$ is small against the sigmoid's voltage scale $1/\rho \approx 1.8$ mV. Human TI polarizations are $\sim 10^{-2}$ mV, over two orders below that, so the quadratic term is the leading nonlinearity throughout the regime this paper treats. Rodent TI is not in that regime: the fields are $\sim 10^3$ times larger, the sigmoid saturates, and the expansion fails. §5 returns to what that costs when rodent results are scaled to humans. Intuitively, the difficulty is that the field carries power only near the carrier $\omega_c$, far from the slow envelope frequency $\Omega$, so some operation must move power down to $\Omega$. The *linear* term cannot: it passes the input lines straight through, reproducing the carrier and its two flanking *sidebands* (the lines at $\omega_c \pm \Omega$ created by the modulation), but still nothing at $\Omega$. The *quadratic* term can: squaring the field multiplies the fast carrier by itself, and because $\cos^2$ has a constant (zero-frequency) part, the squaring folds power down to *baseband*—the low-frequency band near 0 Hz, where the slow envelope lives. This is *rectification*, exactly what a diode does in an AM radio. Keeping the baseband (low-frequency) part of $s^2 = \varepsilon^2(1 + m\cos\Omega t)^2 \cos^2 \omega_c t$ (the fast $\cos^2 \omega_c t$ averages to its mean $\frac{1}{2}$),

$$\tfrac{1}{2}\sigma''\, s^2\big|_{\mathrm{LF}} = \tfrac{1}{4}\sigma''\varepsilon^2\Big[(1 + \tfrac{m^2}{2}) + 2m\cos\Omega t + \tfrac{m^2}{2}\cos 2\Omega t\Big], \tag{9}$$

so a line appears at exactly the envelope frequency $\Omega$, with amplitude

$$A_\Omega \simeq \tfrac{1}{2}\,\sigma''(v^*)\,\varepsilon^2\, m. \tag{10}$$

Physical TI applies two tones rather than the modulated carrier of Eq. (7). The two waveforms are not the same signal: the two-tone sum is suppressed-carrier, with lines only at $\omega_1$ and $\omega_2$, whereas Eq. (7) also carries a line at $\omega_c$. Writing the post-coupling polarizations as $\varepsilon_1$ and $\varepsilon_2$, the same expansion gives $A_\Omega = \frac{1}{2}\sigma''(v^*)\,\varepsilon_1\varepsilon_2$.

What the quadratic term recovers is not the envelope but its square. Writing $A(t)$ for the instantaneous envelope of the two-tone field, $A^2(t) = \varepsilon_1^2 + \varepsilon_2^2 + 2\varepsilon_1\varepsilon_2\cos\Omega t$, the low-pass part of the squared signal is exactly

$$\mathrm{LPF}\big[s^2(t)\big] = \tfrac{1}{2}A^2(t) = \frac{\varepsilon_1^2 + \varepsilon_2^2}{2} + \varepsilon_1\varepsilon_2\cos\Omega t. \tag{11}$$

This is square-law detection in the strict sense: the population transduces $A^2$, not $A$. The distinction is not cosmetic, because $A^2$ is exactly sinusoidal at $\Omega$ whereas $A$ is not—at a matched target ($\varepsilon_1 = \varepsilon_2 = \varepsilon$), $A(t) = 2\varepsilon|\cos(\Omega t/2)|$, a rectified cosine.

The distinction is also spatial. Define the excursion of a periodic quantity as $\mathcal{M}[f] \equiv \max_t f - \min_t f$. The conventional TI envelope map is $\mathcal{M}[A] = 2\min(\varepsilon_1, \varepsilon_2)$, whereas the square-law detector responds to $\mathcal{M}[\frac{1}{2}A^2] = 2\varepsilon_1\varepsilon_2$, so the two are related pointwise by

$$\mathcal{M}\big[\tfrac{1}{2}A^2\big] = \max(\varepsilon_1, \varepsilon_2)\;\mathcal{M}[A]. \tag{12}$$

The square-law map is the conventional envelope-excursion map multiplied, at each point, by the dominant local polarization. It therefore does not preserve conventional TI focality. In typical montages, where the dominant carrier is substantially larger superficially than at depth, this reweighting degrades deep-to-superficial contrast: at $(\varepsilon_1, \varepsilon_2) = (10, 1)$ superficially against $(0.5, 0.5)$ at a matched deep target, $\mathcal{M}[A]$ favors the superficial point by $2\times$ and $\mathcal{M}[\frac{1}{2}A^2]$ by $40\times$. The magnitude of the degradation is not fixed by this algebra and must be evaluated from the actual vector field distribution; a sufficiently weak second carrier can suppress the superficial product instead. The product law is thus not an account of TI focality, and we do not offer it as one.

The selectivity this mechanism does supply is dynamical rather than spatial,

$$R_\Omega(\mathbf{x}) \;\propto\; \sigma''(v^*_{\mathbf{x}})\; G_{\mathbf{x}}(\Omega)\; \varepsilon_1(\mathbf{x})\,\varepsilon_2(\mathbf{x}), \tag{13}$$

a product of three factors. Field overlap sets where a difference-frequency drive is generated at all; the operating point and the network susceptibility set which of the driven populations amplifies it. Resonance does not override an unfavorable field product, but it can compensate for one. This is the radio analogy taken literally: received signal strength times receiver tuning.

Equation (13) is written for scalar projected fields. In general the coupling is vectorial and compartment-specific, $\varepsilon_i(\mathbf{x}) = \boldsymbol{\lambda}{\cdot}\mathbf{E}_i(\mathbf{x})$, so the mixing product $(\boldsymbol{\lambda}{\cdot}\mathbf{E}_1)(\boldsymbol{\lambda}{\cdot}\mathbf{E}_2)$ is signed: its magnitude follows the product of the projected amplitudes, while its phase inverts with either the sign of $\sigma''$ or the relative sign of the two projections. The derivations below take the collinear, scalar case.

The two waveforms agree at $\Omega$ under the quadratic approximation, and on that basis we use the modulated-carrier form for the analysis; they are *not* equivalent beyond it. For matched genuine tones $\mathrm{LPF}[s^2] = \varepsilon^2(1 + \cos\Omega t)$, whereas for the $m = 1$ AM surrogate $\mathrm{LPF}[s^2] = \varepsilon^2(\frac{3}{4} + \cos\Omega t + \frac{1}{4}\cos 2\Omega t)$: the same $\Omega$ amplitude, but a different DC term and a spurious second harmonic at 25% of the fundamental. Since that second harmonic is precisely the observable that distinguishes transduction classes (§2.3), every claim that turns on the waveform is run with genuine two tones: the demodulation demonstration (Fig. 3), the curvature and square-law controls (Fig. 6), the laminar maps (Figs. 7, S6) and the kHz demonstration. The AM surrogate is retained for the dense perturbative sweeps in $(\Omega, p)$ and $(\Omega, C)$, which read only the leading $\Omega$ component, and Fig. 6d verifies that the two agree there.

In the language of [10], the quadratic cross-term $\kappa_2\, s_1 s_2$ between a slow and a fast input implements amplitude modulation/demodulation and the associated intermodulation lines. Demodulation is thus itself an intermodulation product: the line at $\Omega$ is the *difference frequency* of the two applied carriers ($\omega_c$ and $\omega_c+\Omega$), generated by the same quadratic term that—acting a second time, now on an intrinsic rhythm $\omega_0$—produces the $\omega_0 \pm \Omega$ sidebands seen at the spiking level (Fig. S11b). One nonlinear mixer, applied in cascade. Equation (10) carries three predictions, all verified below: the recovered amplitude is quadratic in the drive ($A_\Omega \propto \varepsilon^2$, square-law detection); it follows the sigmoid curvature ($A_\Omega \propto \sigma''(v^*)$, hence zero at the inflection $v^* = v_0$); and, *at fixed post-coupling polarization* $\varepsilon$, it is independent of the carrier $\omega_c$. This last prediction needs care: as a function of the *applied* field the response inherits the coupling $\varepsilon = \kappa(f_c)E_0$, so the field required for a given effect rises with $f_c$ (§2.6)—consistent with the experimental consensus. Carrier independence is thus a statement about the recovered beat component at fixed polarization, not about the applied field.

### 2.3 Transduction class: square law versus rectification

The mathematical envelope of a TI waveform does not by itself determine the physiological response. A detector implements some transduction $F(A)$ of the local envelope $A(t)$, and different detectors implement different $F$. The class matters, because $F$ fixes the spatial map, the harmonic content and the amplitude scaling—all three measurable.

For a narrowband field $s(t) = A(t)\cos\omega_c t$ passing through a memoryless nonlinearity $g$, the slowly varying output is the average over the carrier cycle, $Y_{\mathrm{LF}}(A) = (2\pi)^{-1}\int_0^{2\pi} g(v^* + A\cos\theta)\,\mathrm{d}\theta$. Odd powers average away, leaving only even ones:

$$Y_{\mathrm{LF}}(A) = g(v^*) + \tfrac{1}{4}g''(v^*)A^2 + \tfrac{1}{64}g^{(4)}(v^*)A^4 + \cdots \tag{14}$$

A weak, smooth nonlinearity therefore transduces $A^2$ generically, whatever its detailed shape—which is what the sigmoid mechanism of §2.2 does, and why Eq. (10) is quadratic. Recovering $A$ itself requires a qualitatively different detector: strong rectification. A full-wave rectifier gives $\mathrm{LPF}[\,|A\cos\omega_c t|\,] = (2/\pi)A$ and a half-wave rectifier $(1/\pi)A$, both linear in the envelope, as does a diode-and-capacitor peak detector. Asymmetric channel rectification, threshold crossing and release events are candidate biological realizations, but whether any of them operates at human TI field strengths is open, and we do not assert it.

The two classes are distinguishable at a single site. At a matched target, $A(t) = 2\varepsilon|\cos(\Omega t/2)|$, whose Fourier series $|\cos u| = \frac{2}{\pi} + \frac{4}{\pi}\sum_{n\geq 1}\frac{(-1)^{n+1}}{4n^2-1}\cos 2nu$ gives a harmonic ladder in ratios $A_{2\Omega}/A_\Omega = 1/5$, $A_{3\Omega}/A_\Omega = 3/35$ and $A_{4\Omega}/A_\Omega = 1/21$. The square-law output $\frac{1}{2}A^2 = \varepsilon^2(1+\cos\Omega t)$ contains only DC and $\Omega$. For a mixed detector $F(A) = c_1 A + c_2 A^2$ the pre-network ratio is

$$r_{\mathrm{det}} \equiv \left|\frac{Y_{2\Omega}}{Y_\Omega}\right| = \frac{1/5}{1 + \frac{3\pi}{4}\,c_2\varepsilon/c_1}, \tag{15}$$

so measuring $r_{\mathrm{det}}$ estimates $c_2\varepsilon/c_1$ directly.

Two cautions make this a usable test rather than a trap. First, a smooth detector does generate a second harmonic beyond quadratic order, $Y_{2\Omega}/Y_\Omega \sim (c_4/c_2)\varepsilon^2$, so the square-law signature is a vanishing ratio in the weak-field limit rather than an exact zero. Second, and more seriously, the recurrent network multiplies each harmonic by its own transfer gain, $r_{\mathrm{meas}} = r_{\mathrm{det}}\,|G(2\Omega)/G(\Omega)|$, and a resonant column suppresses the second harmonic hard: driven at $\Omega = \omega_0$ with $\mathcal{Q} = 10$, a true 20% ratio is measured as 0.7%, which would make an envelope follower indistinguishable from a square-law detector. The resonance developed below is thus precisely what would hide the discriminator. The test must therefore be run *off* resonance—at $\Delta f \simeq f_0/4$ both lines sit on the flat part of $G$ and $|G(2\Omega)/G(\Omega)| \approx 1.22$–$1.25$ across $\mathcal{Q} = 3$–$30$—or with $G$ measured independently by a weak tACS sweep and deconvolved.

Sharper still is an amplitude sweep, which tests the scaling rather than one ratio. An envelope-following detector gives $Y_\Omega \propto \varepsilon$ with the harmonic ratio pinned at $1/5$; a weak square-law detector gives $Y_\Omega \propto \varepsilon^2$ with a ratio growing as $\varepsilon^2$. The log–log slope of the envelope-locked response against field amplitude therefore separates the classes without requiring absolute calibration—and the $\varepsilon^2$ control already reported in §4.4 (measured slope 1.99) is that measurement, performed in the model.

### 2.4 Resonance, carrier independence, and the synaptic band-pass

Equation (10) contains no $\omega_c$: the demodulated amplitude is carrier-independent at fixed polarization (§2.2). Under the direct coupling (7) the carrier need only be high enough that the synaptic low-pass suppresses it relative to the envelope, and "inert"—it must not itself fall in, or beat into, the network's responsive band, nor engage other carrier-specific mechanisms (the ion-channel/kHz pathway). Within those bounds the response is flat in $f_c$ (Fig. 5, left). The synaptic suppression is concrete: each JR synapse applies the operator $\hat{L}_s$ with impulse response $h(t) = Aa\,t\,e^{-at}$, i.e. transfer function $H(\omega) = Aa/(i\omega + a)^2$, a critically-damped second-order low-pass with corner $a/2\pi \approx 16$ Hz. It passes the $\sim$10 Hz envelope ($|H| \sim 0.7$) while suppressing a 100 Hz carrier to $\sim$2.5% (Fig. 5, right), so the carrier is filtered out of the loop *after* the sigmoid has used it to demodulate. Thus two distinct mechanisms shape the closed-loop dynamics: the sigmoid generates a squared-envelope component at the beat frequency, whose amplitude is inherently independent of the carrier, and the synaptic low-pass then removes the carrier itself from the spectrum.

What turns this recovered beat component into a large, frequency-selective response is the network. Linearizing the mass about its fixed point gives a leading eigenvalue pair $\lambda_\pm = -\gamma \pm i\omega_0$; near the Hopf the response is dominated by this critical mode, so the whole network behaves as a single weakly damped harmonic oscillator with damping $\gamma$ and natural frequency $\omega_0$ (the gain below is its one-mode, center-manifold reduction). Just below a supercritical Hopf, $\gamma \to 0^+$ and the mass is a sharply tuned (high quality-factor $\mathcal{Q} = \omega_0/2\gamma$) resonator; driving it with the demodulated beat component (10) at $\Omega$ gives the textbook resonance gain

$$G(\Omega) \propto \left[(\omega_0^2 - \Omega^2)^2 + (2\gamma\Omega)^2\right]^{-1/2}, \qquad G(\omega_0) \propto \frac{1}{2\gamma\omega_0}, \tag{16}$$

so the steady response $\sim A_\Omega\, G(\Omega)$ is a Lorentzian peaked at $\Omega = \omega_0$ whose height diverges as the Hopf is approached. Just above the Hopf the mass is an autonomous limit cycle at $\omega_0$, and the demodulated drive *entrains* it (an Arnold tongue), giving a response peak near $\omega_0$ that grows with cycle amplitude.

The resonance amplifies but cannot detect. The universal Hopf normal form, whose nonlinearity is odd, responds to an in-band drive yet produces no line at $\Omega$ at any order in the field (App. B), so detection and amplification are separate requirements rather than two aspects of proximity to criticality.

### 2.5 Operating-point dependence

Because $A_\Omega \propto \sigma''(v^*)$, demodulation is governed by the curvature at the operating point, and in particular by how far $v^*$ sits from the sigmoid inflection (Fig. 6). The Population Excitation–Inhibition Index (PEIX) of [9] is a related but distinct quantity: the sigmoid gain $\sigma'(v^*)$ normalized to its maximum $\sigma'(v_0) = e_0\rho/2$ at the inflection,

$$\mathrm{PEIX}(v^*) \equiv \frac{\sigma'(v^*)}{\sigma'(v_0)} = \mathrm{sech}^2\Big[\tfrac{\rho}{2}\,(v^* - v_0)\Big] \in (0, 1], \tag{17}$$

a dimensionless read-out of where $v^*$ sits, the side given by $\mathrm{sign}(v^* - v_0)$. PEIX thus tracks $\sigma'$ (maximal at the inflection $v_0$), whereas the demodulation gain tracks $\sigma''$ (zero at $v_0$, peaked on the flanks, and sign-reversing across it): the two are complementary, and PEIX peaks exactly where the demodulation gain vanishes. PEIX is therefore a convenient, measurable read-out of the operating point and hence an indirect predictor of $\sigma''$ and of TI efficacy. The mechanism predicts that anything shifting $v^*$—neuromodulation, attention, arousal, or pathology (psychedelics, PV interneuron dysfunction in Alzheimer's disease)—should scale, and can even flip the sign of, the TI response: a falsifiable handle.

### 2.6 Biophysical realization: carrier frequency, fast elements, and state

Carrier independence (at fixed polarization) holds for the direct coupling (7). Real somata are not direct: the membrane (and dendritic cable) low-passes the field before the spike nonlinearity. (The demodulating $\sigma''$ is the *population* transfer; the membrane filter is the upstream *access* stage—this section asks only whether the carrier survives to it.) Modeling this as a first-order filter $H_m(\omega) = 1/(1 + i\omega\tau_m)$, the carrier amplitude reaching the detector is $\varepsilon_{\text{eff}}(f_c) = \varepsilon/\sqrt{1 + (\omega_c\tau_m)^2}$, and since $\Omega \ll \omega_c$ the three AM lines are attenuated almost equally, so

$$A_\Omega(f_c) \;\simeq\; \tfrac{1}{2}\,\sigma''(v^*)\,m\,\varepsilon_{\text{eff}}^2 \;=\; \frac{A_\Omega(0)}{1 + (\omega_c\tau_m)^2} \;\xrightarrow[\omega_c\tau_m \gg 1]{}\; A_\Omega(0)\,\frac{1}{(\omega_c\tau_m)^2}. \tag{18}$$

The demodulated response rolls off as $1/f_c^2$ above the membrane corner (Fig. S4). For a pyramidal soma/dendrite the passive time constant is $\tau_m \sim 10$–$30$ ms (human L2/3 pyramidal cells: $\sim$12 ms [21] and $16.5 \pm 3.7$ ms [22], which bracket the $\sim$16 ms we adopt for the soma, corner $\sim$10 Hz), so the carrier-power factor $|H_m|^2$ is only $\sim 10^{-4}$ at 1 kHz, $\sim 2.5 \times 10^{-5}$ at 2 kHz and $\sim 10^{-6}$ at 10 kHz. Passive somatic demodulation is therefore negligible at TI's kHz carriers—a four- to six-order-of-magnitude suppression.

The escape is that the demodulating nonlinearity need not sit at the soma. At a fast element (axon, axon initial segment (AIS), node of Ranvier; $\tau \approx 0.2$ ms, effective corner $\sim$800 Hz) the carrier is only mildly attenuated in the kHz band ($|H|^2 \approx 0.39$ at 1 kHz, 0.14 at 2 kHz, still $\sim 6 \times 10^{-3}$ at 10 kHz; Fig. S4). A square-law nonlinearity there produces a real envelope-frequency current, and that low-frequency component is *not* attenuated by the membrane—it is exactly what the slow network reads. The per-neuron modulation is small, but two factors make it count: the envelope sits in the band the membrane passes, and the population near a Hopf amplifies it by $G \sim 1/\gamma$ (Eq. (16)). Demodulation at fast elements, resonantly amplified at the population, is thus a route for TI at realistic carriers—consistent with ion-channel rectification at the node [3] but adding network resonance; our direct-coupling demonstration is the $\tau \to 0$ idealization of this limit, and a direct kHz-carrier run (Fig. S8) confirms it.

Because transcranial electrical stimulation (tES)—and TI, a tES with a kHz carrier—is weak and *modulatory*, the fast-element route is a small, sustained bias of the most excitable compartments, not "spiking the AIS at the beat frequency." Since the applied field adds to endogenous drive, a weak bias near threshold shifts spike probability and timing through stochastic resonance [23, 24]. The population sigmoid (2) is itself a mean-field stochastic-resonance device—its smooth slope is the spread of thresholds and noise across the population—so the bias shifts the mean rate through $\sigma'$ while the demodulated beat component emerges through the curvature $\sigma''$.

Both halves are state-dependent—detection through $\sigma''(v^*)$, amplification through $1/\gamma$—so TI and transcranial magnetic stimulation (TMS) efficacy should track brain state, not the stimulus alone. The 100 Hz carrier used throughout is not merely convenient: the membrane attenuation is mild even for moderately fast elements ($|H_m|^2 \approx 0.98$ at $\tau = 0.2$ ms, $\sim$0.1 at $\tau = 5$ ms), so a protocol with $f_c \sim 100$ Hz and $\Delta f$ swept through a region's band is a clean near-term test. The trade-off is spatial: a 100 Hz carrier is less inert off-target than a kHz one (whose subthreshold carrier is the source of TI's focality), so 100 Hz probes the mechanism while kHz optimizes selectivity.

## 3 Methods

### 3.1 Single-column Jansen–Rit: integration, lock-in, and controls

(All symbols and units are collected in App. A, Table 1.) We integrate the single-column JR model of §2.1 (Eqs. (4), parameters in Table 1) as six first-order ODEs. The TI field enters only the pyramidal output sigmoid, Eq. (4), as $s(t)$; the read-out $v = y_1 - y_2$ is *not* fed the field directly, so any power at $\Omega$ in $v$ is produced by the nonlinearity and the recurrent loop, not by trivial pass-through. Unless stated otherwise $f_c = 100$ Hz and $m = 1$.

We integrate with a fixed-step fourth-order Runge–Kutta scheme, $\Delta t = 2 \times 10^{-4}$ s (50 samples per carrier cycle at 100 Hz; reduced to $10^{-4}$ s for carrier sweeps up to 300 Hz). The kHz-carrier runs of Fig. S8 use $\Delta t = 2 \times 10^{-5}$ s for the carrier sweep to 4 kHz (25 samples per carrier cycle at 2 kHz, 12.5 at 4 kHz) and $1.5 \times 10^{-5}$ s for the 2 kHz output spectrum. Halving each step changes the reported lock-in amplitudes by less than 1%. The 10 kHz figures quoted in §2.6 are analytic evaluations of Eq. (18), not simulations. For each measurement the system is settled for 10–16 s (longer near the Hopf, where damping is weak) before a 3–20 s measurement window; the resonance sweeps use a 20 s window so the lock-in averages the limit-cycle beating on the entrainment side. Parameter sweeps are vectorized: the state is an $(N, 6)$ array advanced simultaneously for all $N$ parameter combinations, so a full resonance map runs in seconds. The full bifurcation diagram (Fig. 2) is computed by numerical continuation of the fixed-point and limit-cycle branches. We independently locate the operating supercritical Hopf by solving, for each $p$, the self-consistent scalar equation for the resting fixed point $v = y_1 - y_2$, forming the $6 \times 6$ Jacobian numerically and diagonalizing it: the Hopf is the $p$ at which the leading eigenvalue pair crosses the imaginary axis with $\operatorname{Im}\lambda_+$ in the alpha band. This gives $p \approx 315$ Hz and $f_0 = \operatorname{Im}\lambda_+/2\pi \approx 11.1$ Hz—consistent with the continuation and with the JR bifurcation analysis of [25]. For $p > 315$ the fixed point is a stable focus (resonator); between the SNIC ($p \approx 114$) and the Hopf the column free-runs on the alpha limit cycle.

The response at the envelope frequency is read out with a software lock-in amplifier referenced to $\Omega$—a standard way to isolate the component of a signal at one chosen frequency: multiply by sine and cosine at $\Omega$ and average, so everything *not* at $\Omega$ cancels and only the envelope-locked amplitude survives. Concretely we multiply $v(t)$ by quadrature references and integrate over a Hann-windowed measurement window to obtain the in-phase ($I$) and quadrature ($Q$) components,

$$I = \frac{2}{\sum_k w_k} \sum_k w_k \,(v_k - \bar{v}) \cos \Omega t_k, \quad Q = \frac{2}{\sum_k w_k} \sum_k w_k \,(v_k - \bar{v}) \sin \Omega t_k, \quad A_\Omega = \sqrt{I^2 + Q^2}, \tag{19}$$

with $w$ a Hann window and $\bar{v} = \sum_k w_k v_k / \sum_k w_k$ the windowed mean. For a pure tone $v = A\cos(\Omega t + \phi)$ the product-to-sum identity gives $I \to A\cos\phi$ and $Q \to -A\sin\phi$, so $A_\Omega \to A$ exactly—the factor $2/\sum_k w_k$ is the normalization that returns the physical amplitude, and $A_\Omega$ equals twice the magnitude of the windowed Fourier coefficient at $\Omega$. The one subtlety is that out-of-band components must not leak into the estimate; the dangerous one is the DC level of $v$ ($\bar{v} \sim 8$ mV, far larger than the demodulated signal). Over an exact integer number of $\Omega$-periods

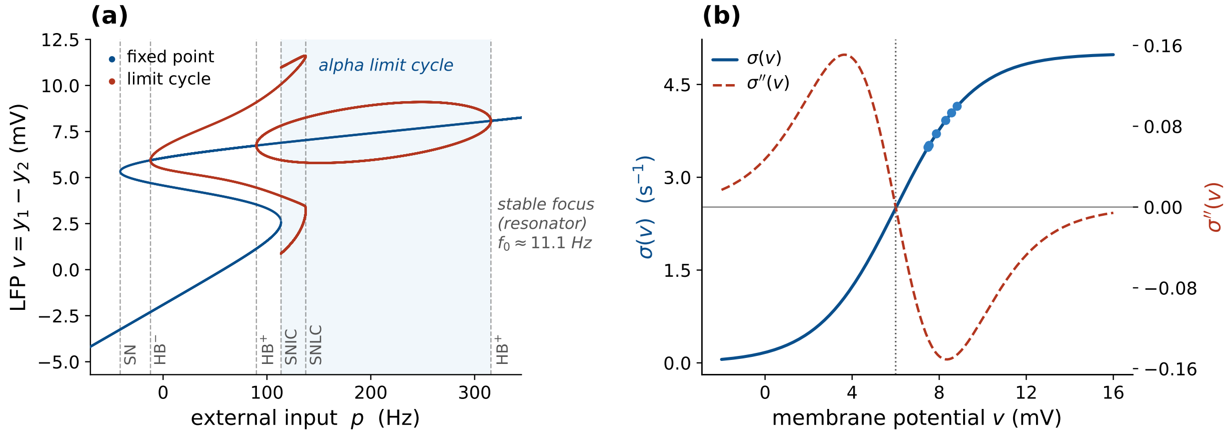


**Figure 2: The Jansen–Rit operating regime: bifurcation and sigmoid curvature.** *(a)* JR bifurcation diagram from numerical continuation—the LFP $v = y_1 - y_2$ along the fixed-point branch (blue) and the min/max of the limit-cycle branches (red) vs. external input $p$, with *solid* stable and *faded* unstable. The alpha limit cycle (shaded) is bounded by a saddle-node-on-invariant-circle (SNIC, $p \approx 114$) and the supercritical Hopf ($p \approx 315$, $f_0 \approx 11.1$ Hz); beyond the Hopf the fixed point is a stable focus—the near-critical resonator regime used throughout. The column is a weakly-damped focus near *both* bifurcations, so it resonates on either approach—more slowly near the SNIC. Further codim-1 points are marked (saddle-node SN, Hopf HB$^{\pm}$, saddle-node of limit cycles SNLC). *(b)* The sigmoid $\sigma(v)$ and its curvature $\sigma''(v)$, which sets the demodulation gain and vanishes at the inflection $v_0$. Dots mark the operating points reached by the closed-loop model for the $p$ values used here.

$\sum_k w_k \cos \Omega t_k = 0$ and there is no leakage, but in general it is nonzero, so we subtract $\bar{v}$ explicitly; this makes the metric exact for a tone and robust down to the $10^{-8}$ level needed for the kHz analysis (§2.6). The signed in-phase component $I$ alone recovers $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ including its sign, which we use to verify the curvature law (Fig. 6).

Three auxiliary constructions support the main results. Throughout we contrast the *open-loop detector*—the firing-rate sigmoid alone at a fixed operating point $v^*$, with no recurrence—against the *closed-loop* column, the full recurrent Jansen–Rit with its feedback. The open-loop detector passes the two-tone field through $\sigma$ at $v^*$ and applies Eq. (19), isolating the pure demodulation law (Figs. 5, 6, S4). The *bifurcation diagram* records the field-free min/max of $v$ vs. $p$, each run initialized at its fixed point plus a small kick. The *two-dimensional map* evaluates $A_\Omega$ over an $(\Omega, p)$ grid. We verify the mechanism three ways: the square law ($A_\Omega$ vs. $\varepsilon$ on a well-damped focus, expecting a log–log slope of 2); a linearization control (replacing $\sigma$ in Eq. (4) by its tangent at $v^*$, which must abolish the $\Omega$ response); and a check of $\sigma', \sigma''$ against finite differences. All code is openly available (see *Code and data availability*).

### 3.2 Laminar two-band model (LaNMM)

We complement the single-column analysis with simulations of the laminar neural mass model (LaNMM) of §2.1. We integrate the full 28-variable column [9, 20], with the field entering exactly as in the single JR column—as a membrane polarization added inside the sigmoid argument of the target population,

$$\varphi_X(t) = \sigma\big(v_X(t) + s(t)\big), \qquad X \in \{P_1, P_2\}, \tag{20}$$

with $s(t)$ the AM or two-tone field of §2.2 (amplitude $\varepsilon$ in mV, beat $\Omega = 2\pi\Delta f$, carrier $\omega_c = 2\pi f_c$) and all three external drives held flat at their baselines (Fig. S1). Coupling the field this way, rather than as a perturbation of an external input *rate*, matters for more than tidiness: a rate cannot go negative, so a rate-coupled AM drive must be clipped at zero once its excursion exceeds the baseline, and a hard clip is a half-wave rectifier—a diode detector placed upstream of the sigmoid, which would inject a line at $\Delta f$ into the drive before the population nonlinearity ever saw it. A signed membrane polarization carries no such constraint and none is applied. Because the LaNMM and JR sigmoids share the voltage scale $1/\rho \approx 1.8$ mV, $\varepsilon$ is directly comparable between the two models. Integration is a vectorized fixed-step RK4 ($\Delta t = 10^{-4}$ s) advancing the whole parameter grid simultaneously, as in the JR sweeps. In place of the lock-in (Eq. (19)) we read the alpha-band (8–12 Hz) power of the $P_1$ and $P_2$ membrane potentials from the Hilbert envelope over a post-transient window—a phase-robust measure suited to the entrainment regime, where lock-in phase slips would otherwise complicate the read-out. Two sweeps probe the column: an amplitude–beat sweep over $(A, \Delta f)$ at fixed carrier (the Arnold tongue), and a carrier–beat sweep over $(f_c, \Delta f)$ at fixed $A$, built on the LaNMM module of [9]. The maps (Figs. 7, S6) show the *stimulation-induced* alpha enhancement, alpha power minus the unstimulated ($A = 0$, no modulation) baseline, which removes the per-simulation offset so the resonant structure is visible. For the alpha resonance we instead drive the deep $P_1$ loop directly and sweep its input against $\Delta f$, reading the lock-in (Eq. (19)) of $v_{P_1}$ at $\Delta f$.

### 3.3 Exact next-generation mean field (NMM2 PING)

We test the mechanism in the exact next-generation mean field (NMM2, §2.1), following the Rosetta Stone formulation of [11] (which places the Montbrió–Pazó–Roxin reduction [17] on a common footing with the JR and LaNMM models used here), where the firing-rate nonlinearity is the derived MPR dynamics of Eq. (3) rather than a postulated sigmoid. We use a two-population excitatory–inhibitory (PING) motif [18, 19, 11]: each population $\alpha \in \{E, I\}$ is an MPR mean field (firing rate $r_\alpha$, mean potential $v_\alpha$) with membrane time $\tau_\alpha$, coupled through second-order synapses,

$$\begin{aligned} \tau_E \dot{r}_E &= \Delta/(\pi\tau_E) + 2r_E v_E, \qquad & \tau_E \dot{v}_E &= v_E^2 + \bar{\eta} - (\pi\tau_E r_E)^2 + \tau_E C\,(s_E - s_I) + \delta V(t), \\ \tau_I \dot{r}_I &= \Delta/(\pi\tau_I) + 2r_I v_I, \qquad & \tau_I \dot{v}_I &= v_I^2 + \bar{\eta} - (\pi\tau_I r_I)^2 + \tau_I C\,(s_E - 2s_I), \end{aligned} \tag{21}$$

with second-order synaptic activations $s_E = \hat{K}_a[r_E]$ (excitatory, time constant $\tau_a$) and $s_I = \hat{K}_g[r_I]$ (inhibitory, $\tau_g$), each obeying the same critically-damped operator as the JR synapse, $\tau^2\ddot{s}+2\tau\dot{s}+s = r$; here $C$ is the global synaptic gain and the weights $(A_{ee}, A_{ei}, A_{ie}, A_{ii}) = (1, 1, 1, 2)$ set the E and I rows—i.e. the $E$ membrane sees $C\,(s_E - s_I)$ and the $I$ membrane $C\,(s_E - 2s_I)$, the only asymmetry being the doubled $I \to I$ weight $A_{ii} = 2$. The common background drive $\bar{\eta}$—the same in both populations, with no separate input current—is the bifurcation parameter: raising it past a supercritical Hopf at $\bar{\eta} \approx 1$ opens a gamma limit cycle (the gamma counterpart of the JR alpha Hopf). The applied field enters the excitatory membrane current as $\delta V(t) = \lambda E(t)$ as two genuine tones at $f_c \mp \Delta f/2$ (carrier $f_c = 300$ Hz, envelope $\Delta f$ swept through the gamma band); the polarization is expressed in the dimensionless MPR units of $v_E$, so the carrier is rectified by the exact quadratic $v_E^2$ term, with no postulated sigmoid. We integrate with fixed-step RK4 ($\Delta t = 0.05$ ms) and read out the lock-in (Eq. (19)) of $r_E$ at $\Delta f$ over the $(\Delta f, \bar{\eta})$ plane. Parameters (Rosetta Stone PING [11]): $\Delta = 1$, global coupling $C = 15$, $\tau_E = 15$, $\tau_I = 7.5$, $\tau_a = 10$, $\tau_g = 2.5$ ms, with $\bar{\eta}$ swept through the supercritical Hopf at $\bar{\eta} \approx 1$ ($f_0 \approx 55$ Hz).

Two parameters reach this gamma Hopf, and we use both as criticality knobs—just as the external drive $p$ and the coupling $C$ both reach the JR alpha Hopf. The background drive $\bar{\eta}$ (above) is swept for the resonance map (Fig. S7); the global coupling $C\,(\equiv J)$ is swept for the J-curve (Fig. S9): we vary $J$ at fixed $\bar{\eta}$ toward the gamma Hopf $J^*$ (located by bisection on the leading Jacobian eigenvalue) and read the lock-in peak of $r_E$ over $\Delta f$. Unlike the JR J-curve, *no operating-point pinning is needed*, because here the rectifier is the exact $v_E^2$ term, whose quadratic coefficient is fixed by construction; sweeping $J$ therefore changes only the damping $\gamma$, isolating the amplification cleanly.

### 3.4 QIF spiking network (microscale of NMM2)

NMM2 is the exact mean field of an all-to-all network of QIF neurons [17, 11, 19], so we integrate that network directly to test the prediction at the spike level (§4.9): $N = 8000$ QIF neurons per population, parameters identical to the PING mean field (Eq. (21)), excitabilities $\eta_j \sim \text{Lorentzian}(\bar{\eta}, \Delta)$ sampled by deterministic quantiles, second-order synapses driven by the empirical population rates, and a genuine two-tone field entering the excitatory membranes. Spiking and threshold dynamics are where the AM surrogate and the physical waveform cannot be assumed equivalent beyond perturbation theory, so this run uses two tones throughout. The bifurcation parameter is the common background drive $\bar{\eta}$, with a gamma Hopf at $\bar{\eta}_{\text{Hopf}} \approx 1.1$ (autonomous $f_0 \approx 53$ Hz; these finite-$N$ values sit just above the exact mean-field Hopf $\bar{\eta} \approx 1$, $f_0 \approx 55$ Hz, the small offset being a finite-size effect); the carrier is $f_c = 300$ Hz and the field amplitude $A_f = 35$. Because this is a *gamma* circuit, the TI beat $\Delta f$ is set in the gamma band: $\Delta f = 55$ Hz *near* the forced gamma resonance ($\bar{\eta} = 0.85$, below the Hopf) and $\Delta f = 42$ Hz in the oscillatory regime ($\bar{\eta} = 1.5$, above the Hopf)—a detuning of $\approx$11 Hz below the autonomous $f_0 \approx 53$ Hz that places the beat outside the 1:1 Arnold tongue, so the above-Hopf response is gating, not entrainment (§C). Envelope-locked timing is quantified by the quadrature lock-in $A_\Omega$ of $r_E(t)$ at $\Delta f$, reported as the modulation depth $A_\Omega/\langle r_E \rangle$ (a stable measure—the off→on lock-in ratio would divide by the near-zero field-free baseline).

### 3.5 Coupling sweeps, timing, and the tACS control (Jansen–Rit)

The coupling sweeps of Figs. 9 and S10, and the transcranial alternating-current stimulation (tACS) control of Fig. S2, use the single JR column with the operating point $v^*$ pinned: for each global connectivity $C$ we solve the field-free fixed-point relation analytically for the external drive $p(C)$ that holds $v^*$ (hence $\sigma''(v^*)$) fixed, vary $C\,(= J)$ toward the alpha Hopf, and read $\gamma, \omega_0$ from the $6 \times 6$ Jacobian. The field is the AM drive of Eq. (1) for TI (Fig. 9; AC lock-in vs DC mean rate for Fig. S10) or a direct in-band sinusoid $s(t) = \varepsilon \cos \Omega t$ ($\varepsilon = 0.1$ mV) for the tACS control (Fig. S2), each read out by the lock-in of $v$ at $\Delta f$.

### 3.6 Vocabulary: forced response, entrainment, and gating

"Entrainment" is used in several senses, so we fix three terms. *Forced response* is the driven, non-autonomous case below the Hopf, quantified by the lock-in amplitude at the beat $\Delta f$ (Eq. (19)). *Entrainment* is reserved for its dynamical-systems meaning, frequency capture of a *self-sustained* oscillator, tested by the 1:1 criterion $|f_{\text{out}} - \Delta f| < 0.3$ Hz on the dominant output frequency [26]. *Gating* is amplitude modulation of a persisting rhythm without frequency capture [27, 28]. The distinction is load-bearing, since capture and gating are dynamically different and the same word would otherwise cover both; Appendix C gives the metrics and the Arnold-tongue protocol.

## 4 Results

We present the evidence on three rungs of the modeling ladder: the single Jansen–Rit column (closed form, analyzable), the laminar LaNMM (two native resonances and genuine carrier dependence), and the exact next-generation mean field (NMM2, the demodulating nonlinearity derived rather than postulated). Each rung shows the same pair—square-law demodulation followed by near-Hopf amplification. Two mesoscale signatures then follow that have no single-cell analog: the network's own coupling sets the gain at fixed detection (the $J$-curve, §4.7), and the resonance amplifies oscillatory *timing* far more than mean *rate* (§4.8)—the spiking-level prediction we test in the QIF network.

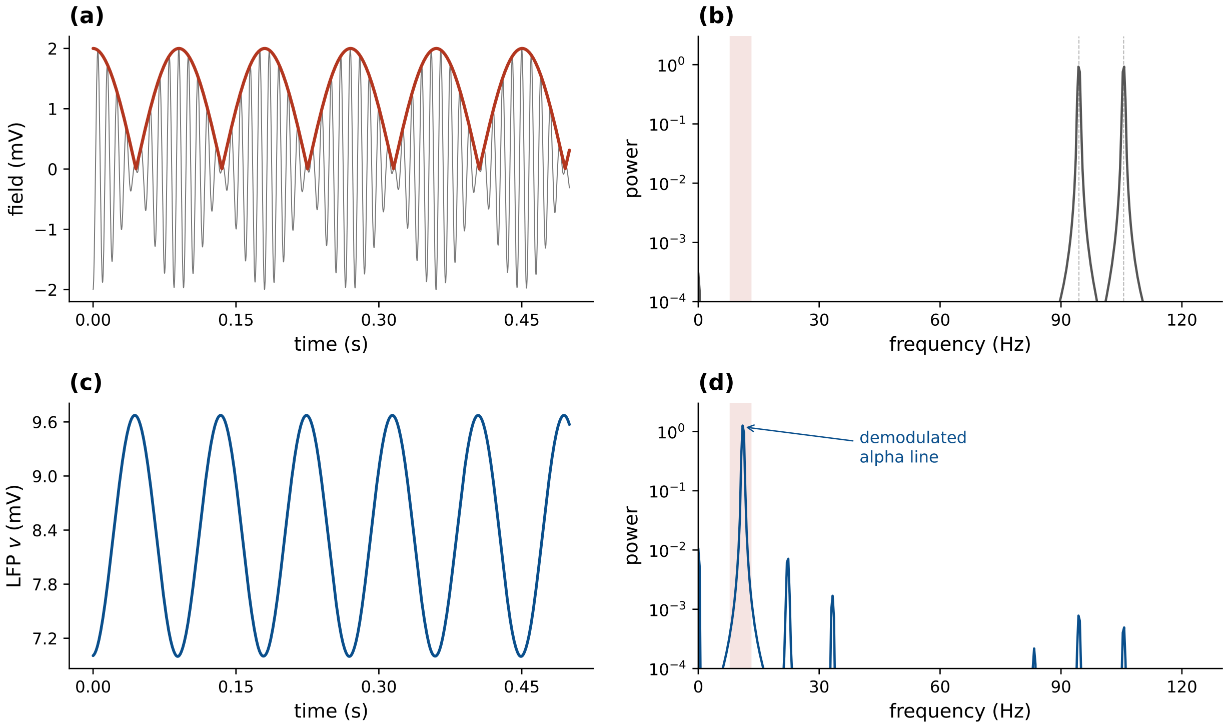


**Figure 3: An alpha rhythm is synthesized from a two-tone TI input (Jansen–Rit below Hopf).** The field is the physical TI waveform, two equal tones at $f_{1,2} = f_c \mp \Delta f/2$, not the amplitude-modulated surrogate. *(a)* The field $s(t)$ and its envelope $A(t) = 2\varepsilon|\cos(\Omega t/2)|$—a rectified cosine, not a sinusoid; it is $A^2$ that is sinusoidal at $\Omega$ (§2.2). *(b)* The input spectrum has lines only at $f_1$ and $f_2$ (94.5 and 105.6 Hz) and no third line at the carrier; power in the alpha band (shaded) is $1.6 \times 10^{-8}$, i.e. empty to numerical precision. *(c)* The output $v = y_1 - y_2$ oscillates at the envelope frequency. *(d)* The output spectrum has a strong peak at $\Omega = 11.0$ Hz, synthesized by the sigmoid and absent from the input.

### 4.1 The Jansen–Rit sigmoid demodulates: an alpha line from a carrier-only input

The system is oriented by Fig. 2: a clean supercritical Hopf at $p \approx 315$ with $f_0 \approx 11.1$ Hz, and a sigmoid whose curvature peaks well off the inflection, so the closed-loop operating points sit where $\sigma''$ is large. The core result is Fig. 3: an input with no spectral power in the alpha band (lines only at $f_c$ and $f_c \pm \Delta f$) produces a clean $\sim$11 Hz output oscillation locked to the envelope, with a strong alpha peak in the output spectrum absent from the input.

### 4.2 Near the Hopf, the demodulated beat drives a sharp resonance

The resonance is quantified in Fig. 4. In the fixed-point regime ($p > 315$, a stable focus) the response is a Lorentzian centered at $f_0$ whose height grows monotonically as the Hopf is approached ($0.23 \to 0.43 \to 0.70$ mV; Fig. 4), i.e. gain $\propto 1/\gamma$. This is genuine forced gain: with no drive the focus is silent at $\Omega$, so all power at $\Omega$ is synthesized by the demodulating nonlinearity and amplified by the recurrent loop. Above the Hopf ($p < 315$) the column free-runs at $f_0$ and the demodulated drive instead phase-locks that autonomous alpha within a 1:1 Arnold tongue; because lock-in amplitude at $\Omega$ then conflates the driven response with the free-running cycle, we characterize that regime with the output frequency $f_{\text{out}}$ rather than the lock-in amplitude (Fig. S3).

### 4.3 Carrier independence and the kHz roll-off

At fixed post-coupling polarization the demodulated response is flat across the carrier band, for both the open-loop detector and the closed-loop column near the Hopf (Fig. 5). As a function of the *applied* field, by contrast, the membrane low-pass makes the response roll off as $1/f_c^2$ above the corner (Fig. S4): passive somatic demodulation is suppressed four to six orders of magnitude

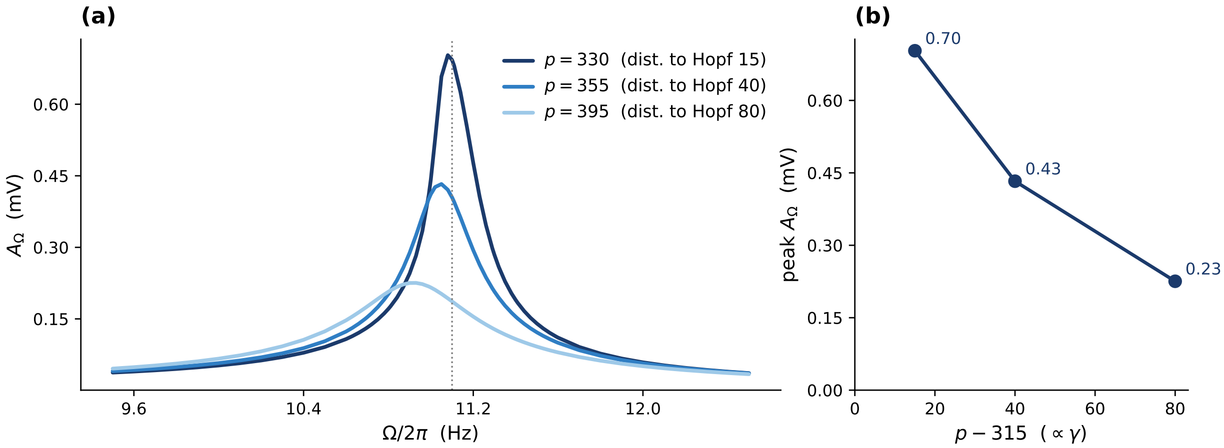


**Figure 4: The demodulated beat drives a resonance that sharpens toward the Hopf (Jansen–Rit).** Fixed-point regime ($p > 315$, stable focus): lock-in response at $\Omega$ vs. envelope frequency ($\varepsilon = 0.3$ mV). *(a)* The demodulated drive forces a Lorentzian resonance at $f_0 \approx 11.1$ Hz whose peak redshifts slightly as the damping grows ($\omega_{\text{peak}} = \sqrt{\omega_0^2 - 2\gamma^2}$; $11.08 \to 10.93$ Hz across the three curves); the response carries no power at $\Omega$ in the absence of drive, so this is genuine forced gain. *(b)* The peak height grows as the Hopf is approached ($p \to 315$, distance to Hopf $\to 0$). Three points cannot establish a power law; the $1/\gamma$ scaling is quantified over a decade, together with its saturation, in Fig. 9(c). The complementary above-Hopf regime, where the demodulated drive *entrains* the autonomous alpha within a 1:1 Arnold tongue, is characterized with the appropriate observable (output frequency $f_{\text{out}}$, not lock-in amplitude) in Fig. S3.

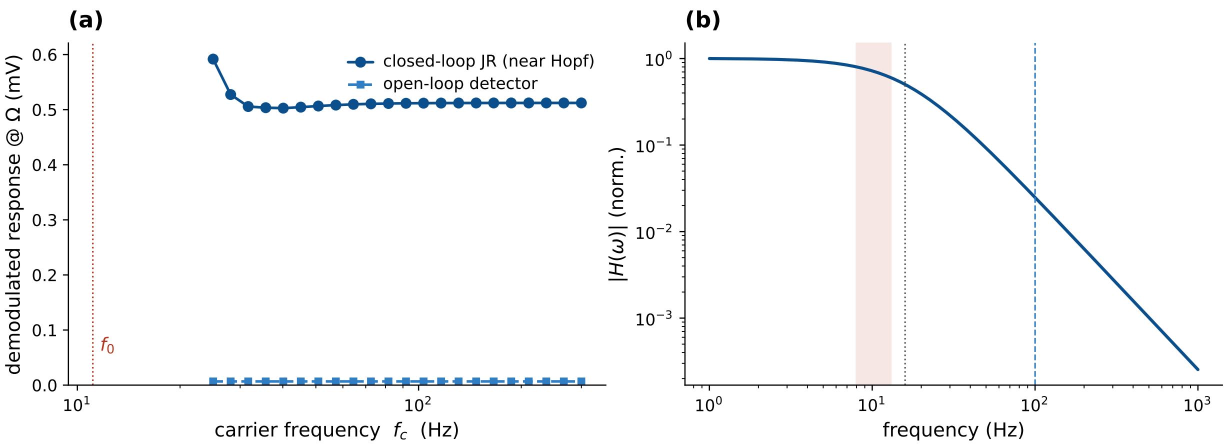


**Figure 5: The demodulated response is carrier-independent at fixed polarization (Jansen–Rit).** *(a)* The demodulated response is flat across carrier frequency $f_c = 25$–$300$ Hz, both for the open-loop detector (exactly, by Eq. (10)) and for the closed-loop JR near the Hopf (which adds $\sim$60$\times$ resonant gain); deviations appear only as $f_c$ approaches the natural band. *(b)* The second-order-synapse band-pass $|H(\omega)|$ passes the envelope ($\sim$0.7) and suppresses the 100 Hz carrier ($\sim$2.5%). Dotted line: synaptic corner $a/2\pi \approx 16$ Hz; shaded band: envelope (8–13 Hz); dashed line: 100 Hz carrier.

at kHz, whereas a fast element ($\tau \approx 0.2$ ms) retains an appreciable response into the kHz band.

To pre-empt the objection that the demonstration runs only at 100 Hz, we drive the full recurrent column with a genuine kHz field, passing it through an explicit membrane low-pass (an extra first-order state, time constant $\tau$) for each coupling route before it reaches the sigmoid (Fig. S8a). The column demodulates and resonates at the recovered alpha well into the kHz band when the nonlinearity sits at a fast element ($\tau \approx 0.2$ ms), exactly as Eq. (18) predicts (Fig. S8a). Driving the fast-element route with two *genuine* tones at $f_{1,2} = 1994.5$ and $2005.5$ Hz—a real TI waveform at a real TI carrier, beating at 10.9 Hz—the input carries $2.2 \times 10^{-13}$ of power in the 8–14 Hz band, i.e. none, while the output carries a clean line at 11.0 Hz (Fig. S5), with the attenuation

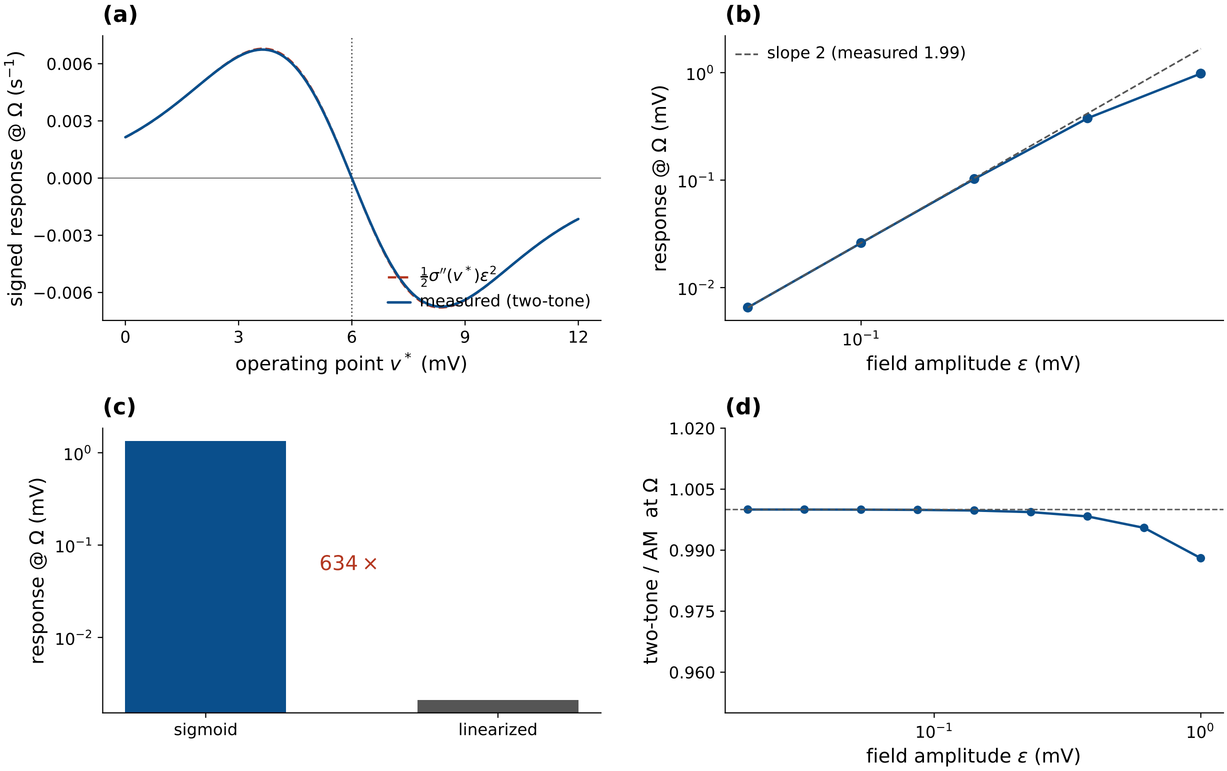


**Figure 6: The curvature law, the square law, and the surrogate control (Jansen–Rit, genuine two tones).** *(a)* The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2$ across the operating range (maximum deviation 0.9%), vanishing at the inflection $v_0$ and reversing sign across it. The mechanism is sign-agnostic: $\sigma''$ sets sign and phase while $|\sigma''|$ sets gain, so the resonance picture is unchanged up to an overall sign in the $\sigma'' > 0$ lobe, which this sweep spans. The sign flip is itself a prediction (§4.10). *(b)* The closed-loop response scales as $\varepsilon^2$, measured slope 1.99. *(c)* Replacing the field-receiving sigmoid by its tangent collapses the response 634×. *(d)* Surrogate control: the two-tone/AM ratio at $\Omega$ approaches 1 as $\varepsilon \to 0$ (1.0000 at 0.02 mV, 0.988 at 1 mV), licensing the AM surrogate for the dense sweeps of Fig. 9, which read only that component, and for nothing else.

set by the analytic $|H_m|^2 = 0.137$ at 2 kHz for $\tau = 0.2$ ms. The kHz claim therefore rests on the physical waveform, not on an extrapolation from the AM surrogate. That the bare unmodulated kHz carrier is itself weakly effective—the premise of TI's focality—is consistent with the limited sensitivity of hippocampal network oscillations to unmodulated kHz fields [29].

### 4.4 The nonlinearity is the detector: the $\sigma''$ curvature law and its controls

The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ across the operating range, vanishing and reversing sign at the inflection $v_0$ (Fig. 6)—demodulation is the sigmoid-curvature term of Eq. (10), and the closed-loop operating points sit where $\sigma''$ is large and negative.

Two controls close the argument (Fig. 6b,c): the response scales as $\varepsilon^2$ at small drive (measured log–log slope 1.99 against the predicted 2), and replacing the field-receiving sigmoid by its linearization collapses the response by 634× (1.34 → 0.0021 mV)—the nonlinearity is the detector, and its curvature is the gain. A third panel settles the surrogate question: at matched amplitude the two-tone and AM responses at $\Omega$ agree to within $10^{-4}$ at $\varepsilon = 0.02$ mV and to 1.2% at 1 mV (Fig. 6d), so the dense perturbative sweeps below, which read only the $\Omega$ component, may use either waveform.

### 4.5 Two native resonances and nonlinear Arnold tongues in the LaNMM

The single JR column isolates and lets us analyze the mechanism; the laminar model (LaNMM, §2.1) confirms it in a cortical column with its *own* two resonances. Driving one population with the two-tone field of Eq. (20) and reading the alpha-band power of the deep ($P_1$, alpha) and superficial ($P_2$, gamma) pyramidal potentials (Methods, Fig. S1), we recover the same envelope resonance, now carrying the column's intrinsic rhythms.

Driving the deep $P_1$ (alpha) loop directly and sweeping the envelope frequency $\Delta f$ reproduces the single-column picture in the laminar model. With the $P_1$ input as the bifurcation parameter, the stable-focus side (high drive) gives a forced alpha resonance at $\sim$10 Hz that grows as the alpha Hopf is approached, while the limit-cycle side (low drive) entrains the autonomous alpha—the same stable-focus/limit-cycle pair as the JR column (Fig. 4), now read out as the deep-pyramidal alpha lock-in. The heuristic laminar model thus shows the JR resonance *and*, in addition, the carrier-dependent Arnold tongues that follow.

One qualification governs the laminar sweeps. They run to $\varepsilon = 3$ mV on $P_2$, above the sigmoid voltage scale $1/\rho \approx 1.8$ mV, so they sit deliberately in a nonlinear, non-perturbative regime. They are a demonstration of tongues and frequency capture, not a verification of the weak-field square law; that verification is the low-amplitude JR and NMM2 work (Fig. 6b), which stays two orders of magnitude below $1/\rho$.

Sweeping the field amplitude $\varepsilon$ and beat frequency $\Delta f$ at a fixed carrier produces *nonlinear Arnold tongues*: alpha power in $P_1$ is maximal when $\Delta f$ matches the deep alpha ($\sim$10 Hz) and grows with $\varepsilon$ (Figs. 7a,b and S6a,b)—envelope resonance and phase locking of the slow generator to the recovered beat component, exactly as in the JR resonance curves but now in a biophysical two-band column.

The carrier dependence reproduces, and physically grounds, the two coupling limits of §2.2–§2.6. Sweeping carrier $f_c$ against beat $\Delta f$: when the field is delivered to the *superficial* loop ($P_2$), strong $P_1$ demodulation requires a *double* resonance—$f_c$ near the $P_2$ gamma ($\approx$44 Hz) *and* $\Delta f$ near the $P_1$ alpha ($\approx$10 Hz) (Fig. 7c,d): the fast loop acts as a tuned "RF front end" that must itself be excited before it can pass a demodulated envelope to $P_1$, and off that gamma resonance the response collapses to $\sim$7% of its peak. When the field is delivered *directly* to the slow generator ($P_1$), the picture inverts: the envelope-locked response persists at every carrier from 25 to 80 Hz, falling only to $\sim$25% of its peak, and the ridge sits at $\Delta f \approx 10.5$ Hz even for a 60 Hz carrier (Fig. S6c)—broadband in $f_c$, as Eq. (10) predicts for direct coupling. This contrast requires the phase-sensitive read-out: because $P_1$ free-runs at its own alpha, band power alone buries a weak envelope-locked drive under an autonomous rhythm that is not phase-locked to the beat, and only the lock-in at $\Delta f$ separates the two (Methods). Real TI, in which the field reaches deep generators through faster elements, sits between these limits: a carrier matched to a fast (gamma) network resonance couples best, but the envelope response persists—weaker—off resonance. This both confirms the JR mechanism and refines it: the "inert carrier" of §2.4 is the direct-coupling idealization, while a band-limited intermediary makes the carrier choice matter, consistent with frequency-tagging/intermodulation paradigms [30].

Concretely the LaNMM predicts that TI efficacy should (i) peak when $\Delta f$ matches a region's intrinsic alpha; (ii) widen and strengthen with modulation amplitude (the tongue); and (iii) for fields delivered via superficial fast circuits, be largest when the carrier is matched to a gamma network resonance—e.g. two nearby gamma carriers summing to a $\sim$40 Hz-centered AM field—while remaining present, weaker, for off-resonance carriers. The third is untested: existing TI studies of deep targets sweep the *beat* frequency rather than the carrier against a network resonance [31, 32], so they neither support nor contradict it.

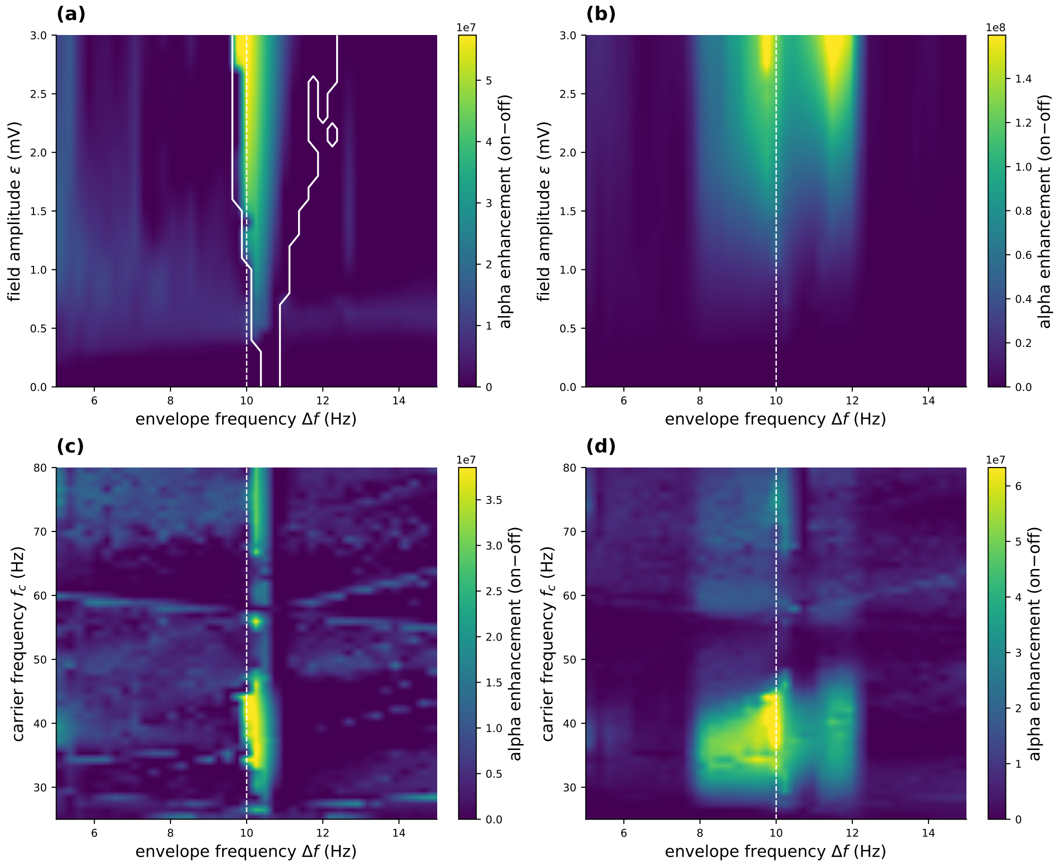


**Figure 7: LaNMM Arnold tongues: two-tone TI field on the superficial $P_2$ (gamma) loop.** The field enters the $P_2$ sigmoid argument as a membrane polarization $\varepsilon$ (mV) with all external drives flat. Color shows alpha-band (8–12 Hz) power minus the unstimulated $\varepsilon = 0$ baseline. *(a,b)* $P_1$ and $P_2$ enhancement vs. $\varepsilon$ and $\Delta f$ (carrier 40 Hz), peaking at the $P_1$ alpha (10.0 Hz, dashed). The white contour in *(a)* is the 1:1 captured set (Appendix C), drawn explicitly because band power alone cannot separate capture from a forced response; its width grows from 0.5 Hz at $\varepsilon \to 0$ to 2.75 Hz at 3 mV. *(c,d)* Enhancement vs. carrier and $\Delta f$ at $\varepsilon = 1.5$ mV: $P_1$ demodulation needs *both* a carrier near the $P_2$ gamma ($\approx$44 Hz) and $\Delta f$ near the $P_1$ alpha—a double resonance.

### 4.6 Exact mean-field confirmation: the NMM2 PING gamma generator

The JR column and the LaNMM both use the heuristic firing-rate sigmoid (NMM1). As a third, independent test we move to the exact next-generation mean field (NMM2, Eq. (3)), where the demodulating nonlinearity is the *derived* quadratic $v^2$ term rather than a postulated curve. We drive the two-population E–I PING motif of Eq. (21) (Methods)—a gamma generator that loses stability through a supercritical Hopf (at $\bar{\eta} \approx 1$, $f_0 \sim 55$ Hz), the gamma counterpart of the JR alpha Hopf—with the two-tone field entering the excitatory current, so the carrier is rectified by the exact $v_E^2$ term.

Driving with a two-tone TI field (carrier $f_c = 300$ Hz, envelope $\Delta f$ swept through the gamma band) and locking in the excitatory rate $r_E$ at $\Delta f$ reproduces the whole picture (Figs. 8, S7). Below the Hopf the response is a forced resonance at the gamma frequency whose height grows as the bifurcation is approached—the $1/\gamma$ gain of Eq. (16); above it the demodulated drive entrains the autonomous gamma. The two-dimensional map traces a ridge at $\Delta f \approx f_0$ that intensifies through the Hopf. Unlike the static-sigmoid models, the resonant frequency itself *shifts with the operating point*—the ridge is diagonal, not vertical—a direct signature of the dynamic, state-dependent transfer of the exact theory [19]. That the same demodulation-and-near-Hopf-resonance mechanism appears with the nonlinearity *derived* from QIF spiking dynamics, rather than assumed, shows it is not an artifact of the heuristic sigmoid.

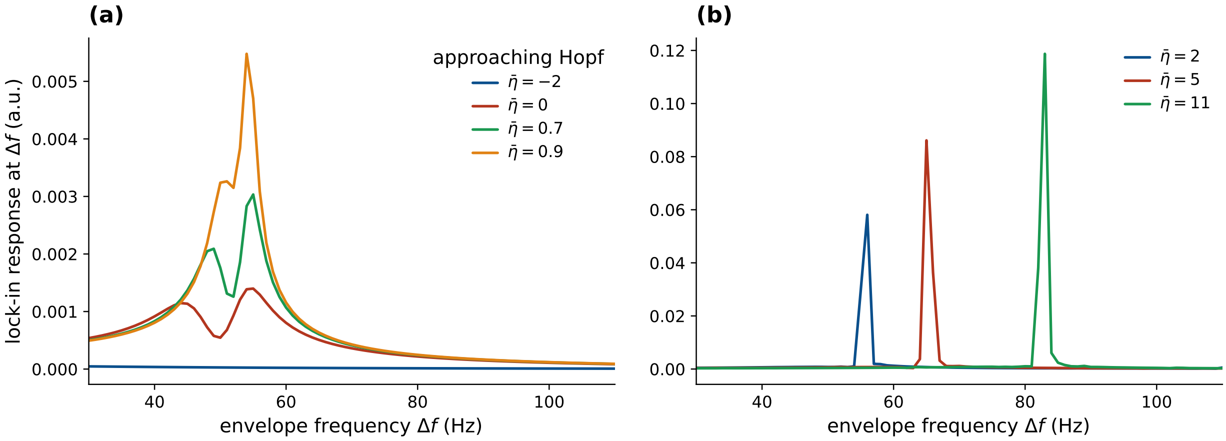


**Figure 8: NMM2 PING gamma resonance (exact mean field, genuine two tones).** Lock-in response of the excitatory rate $r_E$ at the envelope frequency $\Delta f$ under a two-tone TI field (carrier 300 Hz). *(a)* (stable focus, $\bar{\eta} < \bar{\eta}_{\rm Hopf} \approx 1$) forced resonance at the gamma frequency ($f_0 \approx 55$ Hz), the peak growing $0.0014 \to 0.0030 \to 0.0055$ as the Hopf is approached ($1/\gamma$ gain). *(b)* (limit cycle, $\bar{\eta} > \bar{\eta}_{\rm Hopf}$) entrainment of the autonomous gamma, the peak shifting upward with $\bar{\eta}$ ($56 \to 65 \to 83$ Hz). The input carries no power at $\Delta f$; the response is synthesized by the exact quadratic $v^2$ nonlinearity, with no sigmoid anywhere in the model. Matched AM and two-tone drives agree at $\Omega$ to 0.07% well below the Hopf and to 4% at $\bar{\eta} = 0.9$, consistent with the JR surrogate control (Fig. 6d).

### 4.7 What the mesoscale adds: coupling sets the gain at fixed detection

The resonance sweeps above drove the column toward the Hopf with the external input $p$, but $p$ also shifts the operating point, entangling detection ($\sigma''$) with amplification. The sharper, mechanistic statement is that the *network coupling* sets the gain on its own. In the JR column we vary the global connectivity $C$—the coupling strength $J$—and, for each $C$, solve analytically for the drive $p(C)$ that *pins* the operating point $v^*$, so the detection gain $\sigma''(v^*)$ and the open-loop detector amplitude $A_{\rm open} = \frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ are held exactly constant. As $C \to C^*$ (the Hopf, $C^* \approx 137$) the damping $\gamma \to 0$, the demodulated response grows $\sim$30× and the envelope resonance sharpens (Fig. 9a,b)—entirely a network effect, since detection is frozen and only $\gamma$ changes. The recovered gain $A_\Omega/A_{\rm open}$ tracks $1/\gamma$ before saturating near criticality (Fig. 9c); driving the same column to the Hopf through the input $p$ rather than the coupling $C$ yields the same gain–$\gamma$ relation, so the amplification depends only on the distance to criticality, not on which parameter delivers it. All these sweeps stay on the stable-focus side ($\gamma > 0$), approaching the supercritical Hopf from below: the $1/\gamma$ growth is the forced-resonance susceptibility of a damped network, not entrainment of an autonomous cycle. On the other side of the bifurcation the column oscillates on its own, and the demodulated drive instead phase-locks that limit cycle within an Arnold tongue (Fig. S3; also the entrainment side of Fig. 4 and the LaNMM and NMM2 maps)—1:1 frequency capture over a band that widens with drive, quantified by the locking range rather than a diverging $1/\gamma$ gain, the two coinciding only at onset. Because the drive is weak, on the autonomous side it biases phase (timing) more than amplitude, consistent with the timing-not-rate signature (§4.8).

This becomes exact in the next-generation theory. In the NMM2 PING the coupling $C$ is the *literal* mean inter-neuron (QIF) synaptic weight $J$, and the demodulating nonlinearity is the exact $v_E^2$ term, whose quadratic coefficient is a fixed constant—so detection needs no pinning. Sweeping $J$ at fixed background drive $\bar{\eta}$ toward the gamma Hopf $J^*$, the demodulated peak of $r_E$ grows as $1/\gamma$ and saturates at criticality (Fig. S9): the heuristic JR result, reproduced with the coupling *and* the rectifier both derived rather than assumed, and a direct counterpart of the coupling-controlled weak-field sensitivity of QIF networks [19]. The two columns realize one band-agnostic law in their two native bands—alpha (JR) and gamma (NMM2): the single-cell nonlinearity detects the envelope (coarse-grained into $\sigma''$, or the exact $v^2$), while the sharp, tunable gain is the network's—a

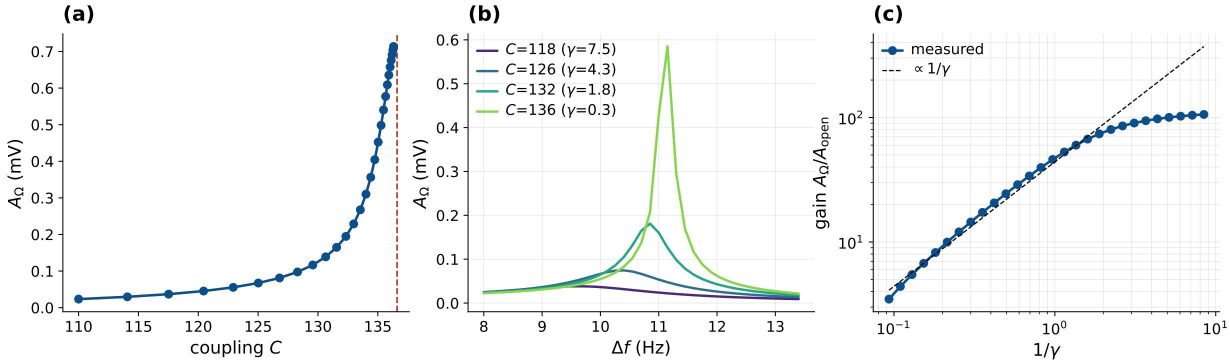


**Figure 9: Coupling controls the amplification at fixed detection (Jansen–Rit).** The operating point $v^*$ is pinned by co-solving $p(C)$, so $\sigma''(v^*) = -0.151$ and $A_{\text{open}}$ are held constant throughout. *(a)* The demodulated $A_\Omega$ grows as the coupling $C\,(= J)$ approaches the Hopf $C^*$ from the stable-focus side ($\gamma > 0$). *(b)* The envelope resonance sharpens and grows as $C \to C^*$. *(c)* The network gain $A_\Omega/A_{\text{open}}$ follows $1/\gamma$ (dashed) and saturates near the bifurcation.

resonant amplification set by proximity to the Hopf and adjustable through the coupling.

### 4.8 The resonance amplifies envelope timing, not the mean rate

The demodulator feeds the network a baseband term and a line at the beat $\Omega$ (Eq. (9)), and the transfer $G(\Omega)$ of Eq. (16) treats them differently: $G(\omega_0) \propto 1/(2\gamma\omega_0)$ diverges as the Hopf is approached, while $G(0) \propto 1/\omega_0^2$ stays flat. Pinning $v^*$ to hold detection fixed, as in Fig. 9, and driving the column toward the Hopf through the coupling, the AC lock-in at the resonant beat $\Delta f = f_0$ grows $\sim$17×, tracking $1/\gamma$ before saturating near criticality, while the DC shift of the mean pyramidal firing rate stays within a few mHz of zero (Fig. S10). The network amplifies the envelope-locked AC response and not the DC rate, so a flat mean rate is the expected signature even where TI is acting—the mesoscopic counterpart of the observation that TI alters spike timing but not firing rate in the primate brain [6].

### 4.9 Microscale confirmation: TI realigns spike timing in the QIF network

The mesoscale prediction has a direct microscale test, since NMM2 is the *exact* mean field of an all-to-all network of quadratic integrate-and-fire (QIF) neurons with Lorentzian-distributed excitabilities [17, 11, 19]. Integrating that spiking network ($N = 8000$ neurons per population, parameters identical to the PING mean field; Methods) shows TI bunching spike *times* into the beat cycle with only a small change in the mean rate, on both sides of the gamma Hopf (Fig. 10). Below the Hopf (forced) the field-free network fires asynchronously at a flat rate; with TI on, spikes condense into envelope-locked bands (Fig. 10a,b). Above the Hopf (oscillatory) the free-running gamma is *gated* by the TI beat—forced cross-frequency modulation (PAC), not frequency entrainment, since the ≈11 Hz detuning lies outside the 1:1 tongue: its bursts group into the $\Delta f = 42$ Hz envelope while the intrinsic $\sim$53 Hz gamma persists (Fig. 10c,d). Because the two rhythms coexist in a nonlinear medium, the sigmoid mixes them into intermodulation sidebands in the population rate—most prominently the direct $\Delta f$ line and the sum component near 95 Hz (53 + 42), with the difference tone near 11 Hz (the detuning itself); this is the spiking-network fingerprint of the same nonlinear frequency mixing that operates at the macroscale, and it is the sideband signature of *gating* rather than the frequency pulling of entrainment. The spiking network reproduces the exact NMM2 mean field (matched gamma spectrum and mean rate, Fig. S11a), and across both regimes the mean rate rises only slightly (×1.07–1.11) while the $\Delta f$ modulation depth of the population rate rises from 0.01–0.03 to 0.17–0.79 (Fig. S11c). The same separation appears per neuron rather than only in the population average: the spike vector strength at $\Delta f$ rises from 0.044 field-free to 0.334 (forced) and 0.159 (gated). Thus envelope-locked timing modulation

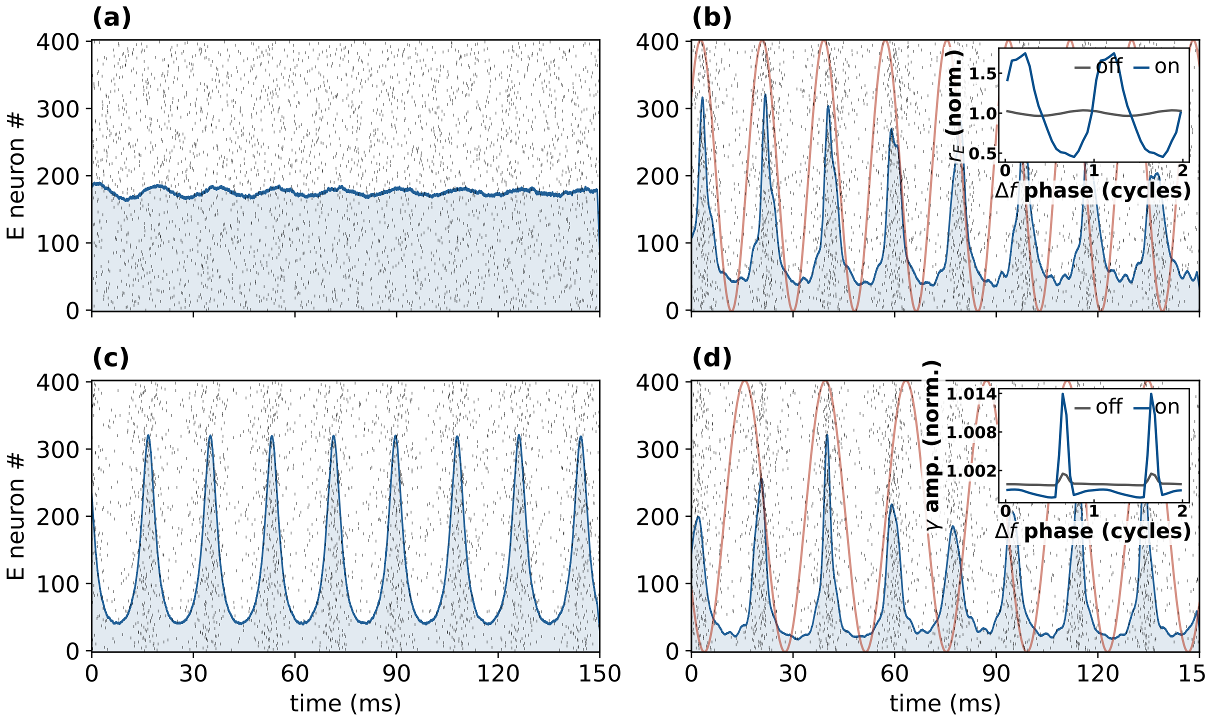


**Figure 10: QIF spiking network: TI realigns spike timing, not mean rate.** $N = 8000$ excitatory and inhibitory quadratic integrate-and-fire neurons whose exact mean field is the NMM2 PING circuit (Eq. (21)). The circuit's intrinsic rhythm is gamma ($\sim$53 Hz), so the beat is set in the gamma range (carrier $f_c = 300$ Hz, genuine two tones). Black ticks are $E$ spikes (400 of 8000), blue the population rate, red the TI envelope; the bifurcation parameter is the common drive $\bar{\eta}$ (gamma Hopf at $\approx$1.1). *(a,b)* Forced ($\bar{\eta} = 0.85$), TI off then on at $\Delta f = 55$ Hz: asynchronous firing becomes envelope-locked bunching. *(c,d)* Oscillatory ($\bar{\eta} = 1.5$), TI off then on with the beat detuned to 42 Hz: the free-running gamma persists but gains a $\Delta f$-locked component. Because the $\approx$11 Hz detuning places the beat outside the 1:1 tongue, this is forced cross-frequency *gating*, not frequency capture (contrast Fig. S3); the detuning is what certifies the locked component as a driven effect. *Insets*, each normalized to its own mean, folded on beat phase (off gray, on blue): *(b)* the population rate; *(d)* the gamma-band *amplitude*, since taking the amplitude removes the gamma carrier so the near-commensurate 53/42 Hz rhythms cannot contaminate the fold. Both are flat with TI off.

dominates rate modulation by roughly an order of magnitude—the spiking-level counterpart of §4.8 and a direct microscale analog of the in-vivo finding that TI alters spike timing far more than firing rate [6].

### 4.10 Falsifiable predictions and tests

The mechanism makes four predictions, each with an experimental handle.

1. **Carrier independence at fixed polarization.** Once the post-coupling polarization $\varepsilon = \kappa(f_c)E_0$ is matched, the leading quadratic beat component no longer depends on $f_c$, provided the carrier lies outside the responsive bands of both detector and network; frequency dependence reappears through the access function $\kappa(f_c)$, or when a fast cellular or network stage is itself resonant (§4.5). The *applied*-field threshold still rises with $f_c$ through the membrane roll-off (Eq. (18))—as single-neuron accounts also predict [4, 3]. *Test:* hold the in-zone polarization fixed (raising $E_0$ to compensate the measured $\kappa(f_c)$) while sweeping $f_c$; the network signature is a flat envelope response. Existing AM-versus-tACS comparisons match the extracellular modulation rather than $\varepsilon$ [6], so this is the correction they lack.
2. **Operating-point (PEIX) control and sign flip.** Because $A_\Omega \propto \sigma''(v^*)$, the response scales with the curvature at the operating point and *reverses sign* across the inflection, vanishing at

$v_0$ where the gain $\sigma'$ (PEIX, Eq. (17)) peaks. *Test:* shift $v^*$ with tonic drive, neuromodulation, or task state and read out the predicted scaling and sign reversal of the TI response.

3. **Transduction class.** A weak smooth nonlinearity transduces $A^2$ and an envelope follower transduces $A$ (§2.3), and the two are separable at a single matched target. The former gives $Y_\Omega \propto \varepsilon^2$ with a second-harmonic ratio vanishing as $\varepsilon \to 0$; the latter gives $Y_\Omega \propto \varepsilon$ with the ratio pinned at 1/5. *Test:* sweep field amplitude off resonance (so that both lines sit on the flat part of $G$) and read the log–log slope together with $Y_{2\Omega}/Y_\Omega$; on resonance the network suppresses the second harmonic by $\sim 3\mathcal{Q}$ and the test must be corrected by an independently measured $G(2\Omega)/G(\Omega)$.

4. **Frequency selectivity and state-dependence.** The response is largest when $\Delta f$ matches a region's resonant rhythm and is amplified as $1/\gamma$ near a bifurcation, so manipulations that move the operating point or the distance to the Hopf—psychedelics, PV-interneuron dysfunction, attention, arousal—should scale, and can abolish or sign-reverse, TI efficacy [9]. *Test:* sweep $\Delta f$ through a region's intrinsic band across brain states and track the resonant peak and its width.

Table 2 confronts these predictions—together with two corollaries the mechanism already shares with published observations—against existing evidence. Two are, in qualitative form, *already supported*: TI alters spike timing but not firing rate in vivo [6] (our $G(\omega_0) \gg G(0)$ asymmetry, §4.8–4.9), and most pyramidal TI responses vanish when synaptic transmission is blocked [7] (a network, not single-cell, read-out). A quantitative reanalysis of these datasets against the model is a natural next step, and §5 specifies the amplitude-exponent fit it would require.

| **Prediction (model basis)** | **Existing evidence** | **Sharp test** |
|---|---|---|
| Transduction class: $Y_\Omega \propto \varepsilon^2$ and vanishing $Y_{2\Omega}/Y_\Omega$ (square law) vs. $\propto \varepsilon$ and 1/5 (envelope following) | — (untested; existing comparisons match extracellular field, not $\varepsilon$) | Amplitude sweep off resonance; transfer-corrected harmonic ratio |
| Timing, not rate: TI amplifies the envelope-locked AC, not the DC mean rate ($G(\omega_0) \gg G(0)$) | *Supported in vivo*: TI alters spike timing but not firing rate, primate [6] | Lock-in of spike timing at $\Delta f$ vs. the mean-rate change |
| Network, not single-cell, read-out (population sigmoid + synaptic recurrence) | *Supported in vitro*: most pyramidal TI responses vanish under synaptic block [7] | Synaptic block should abolish the demodulated response |
| Carrier independence at fixed polarization (no $\omega_c$ in $A_\Omega$) | TI thresholds inherit the carrier's $f_c$-dependence [4, 5]; bare kHz weakly effective [29] | Fix in-zone polarization, sweep $f_c \to$ flat envelope response |
| Operating-point ($\sigma''$) control and sign reversal across $v_0$ | Cell-type-specific TI responses (PV vs. pyramidal) [7] | Shift $v^*$ (neuromodulation/state) $\to$ scaling and sign flip |
| Frequency selectivity, sharpened $\sim 1/\gamma$ near criticality | — (a sharp new prediction; largely untested) | Sweep $\Delta f$ through a region's band across brain states |

**Table 2:** The mechanism's predictions confronted with existing in-vivo / in-vitro evidence. Entries cite documented qualitative findings; quantitative reanalysis is left to future work.

## 5 Discussion

We have given a mesoscopic account of TI in which the firing-rate nonlinearity is a square-law detector and proximity to a Hopf bifurcation turns the recovered beat component into a sharp, frequency-selective response. This complements rather than competes with the single-neuron ion-channel account [3]: both are rectify-then-filter demodulation, but ours is agnostic about which fast cellular nonlinearity supplies the effective curvature, and it inherits the brain's own resonances, so a region's tuning is set by its intrinsic dynamics just as a radio is tuned by its tank. Detection is a population read-out of a coarse-grained single-cell nonlinearity; the amplification is the emergent network property. Beyond contributing to the transfer's shape, the single cell need only supply a fast nonlinearity that samples the carrier.

**Where the carrier-sampling nonlinearity may live.** The literature points to three non-exclusive loci. (i) *Axonal $Na^+$ channels* at nodes and the AIS, where fast voltage-gated $Na^+$ rectifies because activation outpaces inactivation [3, 33] and inactivation-mediated block passes the slow envelope while removing the carrier [4]. (ii) *Presynaptic terminals*, which polarize ∼2–10× more than somata [34, 35, 36] and convert depolarization into release through a steeply nonlinear $Ca^{2+}$ relation. Sensitivity is not speed, however: for an isopotential compartment $\tau_m = R_m C_m$ with both quantities per unit area, so the time constant does not shrink with size—a small bouton has small capacitance and correspondingly large input resistance. A terminal therefore tracks a kHz carrier no better than a soma, and the short axonal time constants come from high-conductance nodal membrane rather than from small size. Terminals are the *sensitive* locus, not the fast one, and the carrier sampling is supplied by (i). (iii) *Threshold spiking*, where a weak bias near threshold is rectified through stochastic resonance [24, 23]. The same logic applies to TMS, whose pulses carry kHz content that a 16 ms soma attenuates by ∼50 dB, so it too must act through fast elements whose chronaxie sets a corner near ∼800 Hz [37, 38].

**The conventional TI dose map is an assumption about physiology, not a consequence of field physics.** The envelope is a mathematical property of a high-frequency waveform. It is not itself a low-frequency field, and nothing in Maxwell's equations delivers it to a neuron; biology must construct it. Targeting work since [1] takes the envelope-modulation amplitude $\mathcal{M}[A] = 2\min(\varepsilon_1, \varepsilon_2)$ as the spatial dose, which assumes $R(\mathbf{x}) = \Phi(\mathcal{M}[A])$ for monotonic $\Phi$—that the response depends on the envelope excursion *alone*, and not separately on the carrier baseline, the operating point or the orientation. That assumption is substantive, and a weak smooth nonlinearity cannot satisfy it: the carrier average Eq. (14) is invariant under $A \to -A$, so its expansion contains only even powers and no term proportional to $A$ exists. Square-law transduction is therefore the generic perturbative answer and envelope following is the special case needing justification, since recovering $A$ requires rectification (§2.3) and any *smoothed* threshold reverts to $A^2$ as the field weakens.

We doubt there is a single universal TI mechanism. Near threshold a sharp channel or spike nonlinearity may act as a rectifier and approximate envelope following; well below it, the leading response of any smooth, memoryless cellular nonlinearity is square-law mixing, filtered and amplified by the network as described here. Which of the two operates at human field strengths is an empirical question that present evidence does not settle. Primate recordings at human-comparable fields find an AM waveform less effective than matched-depth tACS, but do not estimate the difference closely enough to select a transduction law [6], and peripheral-nerve work argues that some TI responses reflect carrier integration rather than envelope extraction [5]. What settles it is not whether TI works but which quantity it transduces, and §2.3 gives the measurement.

**The amplifier is not specific to TI.** A tACS field oscillates in the envelope band directly and needs no demodulation, but it drives the same near-Hopf resonator. Sweeping the coupling toward the alpha Hopf at *pinned* operating point, a direct $\Delta f$ drive reproduces the same $1/\gamma$ growth and resonance sharpening as the demodulated TI drive, linear in the field rather than square-law (Fig. S2). That coupling controls this gain is not itself new: the dynamical transfer function of an E–I network is known to become resonant near a Hopf bifurcation, with a peak that *diverges* on the bifurcation line as connectivity is varied [39], and increased self-connectivity is known to enhance weak-field resonant sensitivity in the exact mean field [19]. What the pinned sweep adds is the separation of amplification from *detection.* Holding $\sigma''(v^*)$ exactly constant while only $\gamma$ varies isolates the network gain, and shows that the same amplifier acts on a drive the population has itself demodulated, which is why it cancels when TI and tACS are compared. TI then differs from tACS only by a demodulating front end that converts a kHz carrier's envelope into an effective in-band drive; downstream, both share the criticality- and coupling-controlled amplifier, so—matched for effective in-band drive—they should entrain alike.

**Amplification is critical slowing, of a mean-field kind.** Gain $\propto 1/\gamma$ as the damping vanishes at the Hopf is the dynamical signature of *critical slowing*: a susceptibility that diverges as a system approaches a bifurcation. Whether that warrants the word "criticality" depends on the model. In the heuristic Jansen–Rit and LaNMM columns the Hopf is a bifurcation of a finite-dimensional, fitted ODE, a near-bifurcation statement rather than a phase transition. In the exact next-generation mean field it is sharper: the reduction from the spiking QIF population is exact in the thermodynamic limit and $(r, v)$ are genuine *order parameters*, conformally related to the Kuramoto–Daido order parameter through the Ott–Antonsen ansatz [40, 17], so the Hopf is a true macroscopic transition and the $1/\gamma$ gain its diverging susceptibility. One qualifier keeps this honest. Because the reduction assumes all-to-all coupling the critical point is *mean-field*: it delivers temporal critical slowing but no diverging *spatial* correlation length, so the column realizes the temporal-susceptibility face of criticality and not the scale-free spatial one. With $J$ as the control parameter (Fig. S9), TI drives a population toward a genuine, if mean-field, critical point and reads out its susceptibility.

**Prior work already places the operative nonlinearity in the network.** Recurrent excitation, with short-term synaptic depression, accounts for tACS entrainment in a large-scale spiking model and yields an Arnold tongue over stimulation amplitude and frequency [41]; the dynamical transfer function of an E–I network, in both rate and spiking form, is resonant near a Hopf bifurcation with a peak that diverges on the bifurcation line as the connectivity is swept [39]; an amplitude-modulated high-frequency drive phase-locks a cortical model to its envelope through the same target engagement as low-frequency tACS [42]; and comparing neuron models directly, integrate-and-fire membranes fail to reproduce observed TI behavior while Hodgkin–Huxley ones succeed [43]. The general case for stimulation as interaction with ongoing dynamics has been made explicitly [44]. What we add is the pair of factors—$\sigma''$ as the detector in closed form, $1/\gamma$ as the amplification tied to a measurable relaxation time—and, downstream of both, that the transduced quantity is $A^2$ rather than $A$, a distinction the cited network studies do not draw.

**The amplitude budget does not close at human field strengths.** Two hurdles stand between a transcranial field and a response at $\Delta f$, and they need different answers. The first is that weak fields barely move a membrane: measured coupling is 0.12–0.21 mV per V/m in hippocampus [45, 46] and up to ∼0.5 mV per V/m for cortical L5/6 pyramidal cells [47], human cortex sees about 0.8 V/m at 2 mA [48, 49], and most applied current is shunted through scalp and skull [50]. The field shifts a soma by a few tenths of a millivolt against the ∼23 mV separating rest from threshold [51]. This is as true of tACS as of TI. The second hurdle belongs to TI alone: its field carries no power at $\Delta f$, so the response there must be manufactured, and that is what $\sigma''$ does. The two answers do not substitute for each other. Resonance clears the first—a population near its own bifurcation responds to a weak push far more strongly than an isolated cell—but not the second, because it amplifies TI and tACS alike and cancels when the two are compared.

The resonance buys a factor of order ten. With $\tau = 1/\gamma$ the reverberation time, the gain at resonance relative to the same circuit's response to a steady input is $\mathcal{Q} = \pi f_0 \tau$. Cortical population activity relaxes over a few hundred milliseconds [52], giving $\mathcal{Q}$ of order 10 at $f_0 = 10$ Hz—an order of magnitude, not a value, since that timescale is wideband rather than alpha-specific. Gain also costs bandwidth: the resonance is $1/(\pi\tau)$ wide, about 1 Hz, so a $\Delta f$ fixed at a group-mean alpha will drift outside it during a session. At equal field strength TI is far weaker than tACS. With per-carrier polarizations $\varepsilon_{1,2}$ the demodulated drive is $\frac{1}{2}\sigma''\varepsilon_1\varepsilon_2$ against $2\sigma'\varepsilon$ for tACS at the same peak polarization, a ratio $\eta = \sigma''\varepsilon/4\sigma'$ proportional to $\varepsilon$: a square-law detector gets *less* efficient as the signal weakens. Formally $\eta$ reaches unity near $\varepsilon \approx 7$ mV, but that is the order of the distance to threshold and lies outside the expansion's validity, so parity is not reachable by this route rather than merely expensive. At 1 V/m with a 1 kHz carrier—now comparing at matched *applied* field, so the carrier's membrane attenuation enters twice while the in-band tACS drive is unfiltered—TI comes out $\sim$300$\times$ weaker than tACS if the carrier reaches a fast nodal compartment, and $\sim 10^6\times$ weaker if it reaches only the soma.

The one in-regime measurement sits well above this estimate. Vieira et al. worked at modulation depths of 0.5–0.62 V/m, chosen deliberately to match human studies rather than the 60–383 V/m of the rodent literature, and report TI at 0.34 times the strength of conventional tACS (95% CI 0.04–0.65), or $\sim$12% per unit current under their stated assumption that effects are roughly proportional to applied current [6]. That normalization is not neutral here, since a square-law drive scales as $I^2$ while tACS scales as $I$; the per-current figure is therefore transduction-class-dependent and we do not use it. Their AM-versus-tACS comparison is the most direct available. Having measured that transcranial current is shunted increasingly with frequency, they compensate for it and deliver conventional tACS and an AM waveform through the same electrodes, with the modulation depth near the recorded cells calibrated to 1.14–1.17 times the tACS amplitude. Against a baseline PLV of 0.043, tACS raises the median to 0.20 (95% CI 0.043–0.64) and AM-tACS to 0.10 (0.023–0.38), a paired difference significant at $p = 0.01$.

Taking their fields with a favorable cortical coupling ($\sim$0.5 mV per V/m) through a fast element at 2 kHz gives $\varepsilon \approx 0.1$ mV and hence $\eta \approx 0.8\%$, far below the ordering they report; a somatic route is worse by two further orders. Three limits keep this from being a quantitative test. The comparison rests on twenty neurons, of which ten responded to at least one modality and seven to both, so the sample is selected for responsiveness and the reported intervals are correspondingly broad. PLV is a downstream statistic that depends on baseline locking, network susceptibility and state as well as on the generated drive, and in their full dataset baseline PLV predicts the sign as well as the size of the change. And the quantity matched was the extracellular modulation, not the post-coupling polarization $\varepsilon = \kappa(f_c)E$ of Eq. (8) at whichever compartment supplies the nonlinearity, so the ratio reports the whole end-to-end path from applied field through frequency-dependent access, transduction, network dynamics and spike-phase statistics. It does not isolate the transducer. What the experiment establishes is that a high-frequency AM waveform drives phase locking less effectively than a matched-depth low-frequency one; the magnitude of that difference is not estimated precisely enough to discriminate among candidate transduction laws.

The obvious escape is the operating point, since both $\sigma''$ and $\sigma'$ depend on it and the column might sit somewhere far more favorable. It does not escape. For the logistic transfer, writing $s = \sigma/2e_0 \in (0, 1)$, we have $\sigma' = 2e_0\rho\, s(1 - s)$ and $\sigma'' = 2e_0\rho^2 s(1 - s)(1 - 2s)$, so $\sigma''/\sigma' = \rho(1 - 2s)$ and, since $\mathrm{PEIX} = 4s(1 - s)$ gives $(1 - 2s)^2 = 1 - \mathrm{PEIX}$,

$$\eta = \frac{\rho\,\varepsilon}{4}\sqrt{1 - \mathrm{PEIX}} \;\le\; \frac{\rho\,\varepsilon}{4}. \tag{22}$$

The efficiency is therefore state-dependent exactly as the mechanism predicts—maximal where the population is saturated and far from the inflection, and vanishing at $\mathrm{PEIX} = 1$, where the transfer is locally linear and cannot demodulate at all—but at fixed $\rho$ it is *bounded* by $\rho\varepsilon/4$ whatever the state. The maximum is one of ratio rather than of absolute response: deep in the saturated tails $\sigma'$ and $\sigma''$ both vanish and the TI and tACS drives disappear together. At the physiological Jansen–Rit slope ($\rho = 0.56$ mV$^{-1}$, transfer width $1/\rho \approx 1.8$ mV) that ceiling is 1.4% at Vieira et al.'s fields and 0.28% at human ones. Brain state moves the efficiency, but at fixed $\rho$ it cannot lift the transfer off that ceiling.

The slope is itself a parameter, so the question is what the whole sigmoid family can do. The logistic transfer depends on voltage only through $\rho(v - v_0)$, so writing

$$u = \rho\,\varepsilon, \qquad x = \rho\,(v^* - v_0), \tag{23}$$

we have $\sigma(v^* + \varepsilon q(t)) = 2e_0\, L(x + u\, q(t))$ for any normalized waveform $q$, with $L(z) = (1 + e^{-z})^{-1}$. Every Fourier amplitude, and hence their ratio, is a function of $(u, x)$ alone; the factor $2e_0$ cancels. Maximizing over $v^*$ and $\rho$ at fixed $\varepsilon$ is therefore the same as maximizing over $(u, x)$, and the family supremum is independent of field strength. Evaluating the exact cycle response rather than its quadratic truncation and optimizing $x$ at each $u$, the ratio rises monotonically over the range explored and converges to the hard-threshold limit

$$\sup_{\rho,\, v^*} \eta = 0.3479, \tag{24}$$

approached as $\rho \to \infty$ at a threshold offset $\delta/2\varepsilon = 0.523$. There is no finite optimal transfer width: narrower is always better, and reaching 95% of the ceiling requires $u \gtrsim 6$, that is $1/\rho \lesssim \varepsilon/6$.

Two features of that limit decide the reading. Such a detector is not weak. At the optimum, and at the Vieira-scale polarization $\varepsilon \simeq 0.1$ mV, its open-loop beat drive exceeds the physiological-slope sigmoid's by some three orders of magnitude for the same $e_0$ and waveform, because the field excursion is about twice the threshold offset and the transfer switches across its full range instead of responding linearly. The disparity grows as the field weakens, the smooth sigmoid's response falling as $\varepsilon^2$ while a threshold repositioned with the field keeps its excursion. The obstacle is the width. At Vieira et al.'s fields the ceiling is approached only for $1/\rho \lesssim 17$ $\mu$V, and at human polarizations for $1/\rho \lesssim 3$ $\mu$V, with the threshold offset held to the same order as the applied polarization. The width of a neural-mass sigmoid is set by the spread of thresholds and noise across $\sim 10^4$ neurons and is of order millivolts. A microvolt width lies far below that, and reads more naturally as an idealized microscopic threshold or rectifier than as a neural-mass sigmoid; we do not offer it as a measured single-cell property, real thresholds being themselves noisy and state-dependent. The regime in which the family approaches its ceiling is thus one in which the "population sigmoid" has ceased to describe a population.

A sigmoid steep enough to approach the ceiling is no longer detecting through curvature but rectifying, so the two transduction classes of §2.3 are limits of one family rather than rival models. Which limit operates in vivo is not settled by the evidence available. The ceiling (24) should not be compared statistically with any particular reported response ratio: the numerical proximity of the two would be neither support nor exclusion, for the reasons set out above. Vieira et al. read their own data as favoring rectification, invoking the radio analogy explicitly and proposing asymmetry between inward $Na^+$ and outward $K^+$ channels as the operative nonlinearity. They also note that circuit-level frequency dependence could contribute to demodulation, which is the mechanism developed here.

A retrospective test is available in the existing primate data. Fitting a modality-specific response law $\Delta\mathrm{PLV} = \Psi(aI^p)$, with baseline PLV and state as covariates, and estimating the amplitude exponent $p$ separately for conventional and AM stimulation, tests $p \approx 1$ against $p \approx 2$ on recordings already made. It requires within-modality amplitude variation across comparable cells, an independent estimate of the frequency-dependent access $\kappa(f_c)$, and no selection on responsiveness. Prospectively, matching post-coupling polarization rather than extracellular field alone, and reading the amplitude slope together with the transfer-corrected harmonic ratio of §2.3, separates the classes without an absolute calibration.

None of this touches the amplification half, and it is worth being precise about why. The $J$-curve reports $A_\Omega/A_{\mathrm{open}}$, the closed-loop response normalized by the open-loop detector output at the same operating point. The detector divides out by construction: whatever manufactures the beat component—a smooth population curvature, a nodal rectifier, a spike threshold—the recurrent network's response to that in-band drive is the same, and it is that response which follows $1/\gamma$ and sharpens toward the bifurcation. The same holds in the exact mean field, where the rectifier is the derived $v_E^2$ term and no operating-point pinning is needed at all. Detection and amplification are therefore not merely conceptually separable but experimentally separable, and today's evidence bears only on the first. The population contribution established here is the tuning and amplification; detection may occur microscopically or through the effective population transfer.

We therefore do not claim that square-law transduction is what operates in the human brain at safe field strengths. What the parity argument establishes is narrower and still holds: *any* smooth weak nonlinearity must transduce $A^2$, so a square-law route exists and is the generic leading perturbative form rather than a quantitative estimate of the in-vivo response. The evidence available is consistent with a more rectifying transduction dominating in vivo, in which case the population sigmoid is not the detector and the single cell is, but it does not establish that. The amplification half is unaffected either way, being waveform-agnostic: whatever manufactures the beat component, a network near its bifurcation amplifies it as $1/\gamma$. Deciding between the two routes is exactly what the amplitude sweep of §2.3 does, and it is the measurement we would most like to see: a slope of 1 with a harmonic ladder at 1/5 confirms rectification, a slope of 2 with a vanishing ladder confirms the square law.

**Limitations.** The field enters the population transfer directly; we treat the field-to-nonlinearity coupling $\kappa(f_c)$ phenomenologically rather than modeling channel kinetics, and $\varepsilon$ should be read as post-coupling polarization. The population sigmoid is a coarse-grained stand-in for distributed single-cell nonlinearities. JR alpha near this Hopf is high quality-factor, so the stable-side peaks are narrow and would be broadened by noise and population heterogeneity; the columns here are single, deterministic and noiseless, and coupled noisy columns with an explicit $\kappa(f_c)$ remain the natural next step. Cascading ordinary neural-mass transfers does not solve carrier access, since the intervening synaptic filter removes the kHz component before a second sigmoid can rectify it. A population of microscopic rectifiers could supply the effective detector, but heterogeneity in their thresholds would broaden the aggregate transfer, and whether it stays sharp enough at human fields is a separate biophysical question. The human evidence is also not uniformly positive: a controlled study found peripheral but no central effects [53], and a systematic review reports that occipito-parietal TI did not modulate alpha power [54]. The narrowband prediction turns the second into a design question—whether $\Delta f$ tracked individual alpha closely enough to stay inside a $\sim$1 Hz response band—which is a prediction to be tested, not an explanation to be assumed.

## 6 Conclusion

In the weak-field limit a smooth memoryless transfer—and more generally any smooth detector with a nonzero second-order Volterra kernel—transduces the *square* of the TI envelope rather than the envelope itself: the carrier average contains only even powers, so the leading response is $\frac{1}{4}g''(v^*)A^2$ and there is no term linear in $A$. Recovering $A$ would require rectification, and any smoothed threshold reverts to $A^2$ as the field weakens. The population sigmoid is thus not an exotic detector but the generic perturbative one, and the beat component it generates, $\frac{1}{2}\sigma''(v^*)\varepsilon_1\varepsilon_2$, is what a second-order-synapse network near a Hopf then amplifies as $1/\gamma$. Its efficiency against a direct in-band drive is bounded, however, and the bound is approached only where the transfer has narrowed to a single-cell scale, so whether the population or the cell detects is left open here. A fast single-cell nonlinearity—$Na^+$ at nodes or the AIS, persistent $Na^+$—need only deliver the carrier to that transfer, with presynaptic terminals contributing enhanced polarization and nonlinear release downstream rather than carrier sampling; the mechanism is agnostic about which supplies it.

Both halves are state-dependent, detection through $\sigma''(v^*)$ and amplification through $\gamma$, so TI efficacy is as much a property of brain state as of the stimulus: largest near a bifurcation, tunable by connectivity at fixed detection, and reversible in sign by shifting the operating point. That the same amplifier serves a direct in-band drive means tACS inherits it too, which is why resonance cannot recover TI's square-law penalty.

If the transduced quantity is $A^2$ rather than $A$, the mechanism predicts a spatial response map that is the conventional envelope map reweighted by the dominant local polarization, a response quadratic rather than linear in field amplitude, and an envelope-locked spectrum without the harmonic ladder an envelope follower would produce. These are measurable, and they are what would settle which quantity TI actually transduces.

## Code and data availability

All simulation and figure-generation code, the committed datasets, and a one-command reproduce script (`run_all.py`) are openly available at `https://github.com/giulioruffini/TI-demodulation-paper`, archived at Zenodo (doi:10.5281/zenodo.21009618). The repository `README` documents the per-figure script mapping (a figure→script table) and the dependency order, so every figure in this note can be regenerated from the committed inputs. The code is released under CC-BY-4.0.

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## Competing interests

GR is Chief Scientific and Technology Officer of Neuroelectrics, which develops and commercializes transcranial stimulation devices. BM, AJ and FC are affiliated with Neuroelectrics. The remaining authors declare no competing interests.

## Author contributions

Using the CRediT taxonomy: **G. Ruffini** — conceptualization, methodology, formal analysis, software, investigation, visualization, writing (original draft), supervision, funding acquisition. **B. Mercadal** — validation, visualization, writing (review and editing). **A. Just** — software, visualization, writing (review and editing). **R. de Palma Aristides** — software, visualization, writing (review and editing). **F. Castaldo** — conceptualization, writing (review and editing). **S. Canals** — supervision, writing (review and editing). **C. Mirasso** — supervision, writing (review and editing). All authors read and approved the final manuscript.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Anthropic Claude and OpenAI ChatGPT to streamline the narrative and to assist with code and figure generation from data the authors produced. The authors reviewed and edited all resulting content and take full responsibility for the content of the published article.

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